Learning Physics from Data with Imperfect Models

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Wayne State University

Heavy-Ion Collision Phenomenology Journal Club

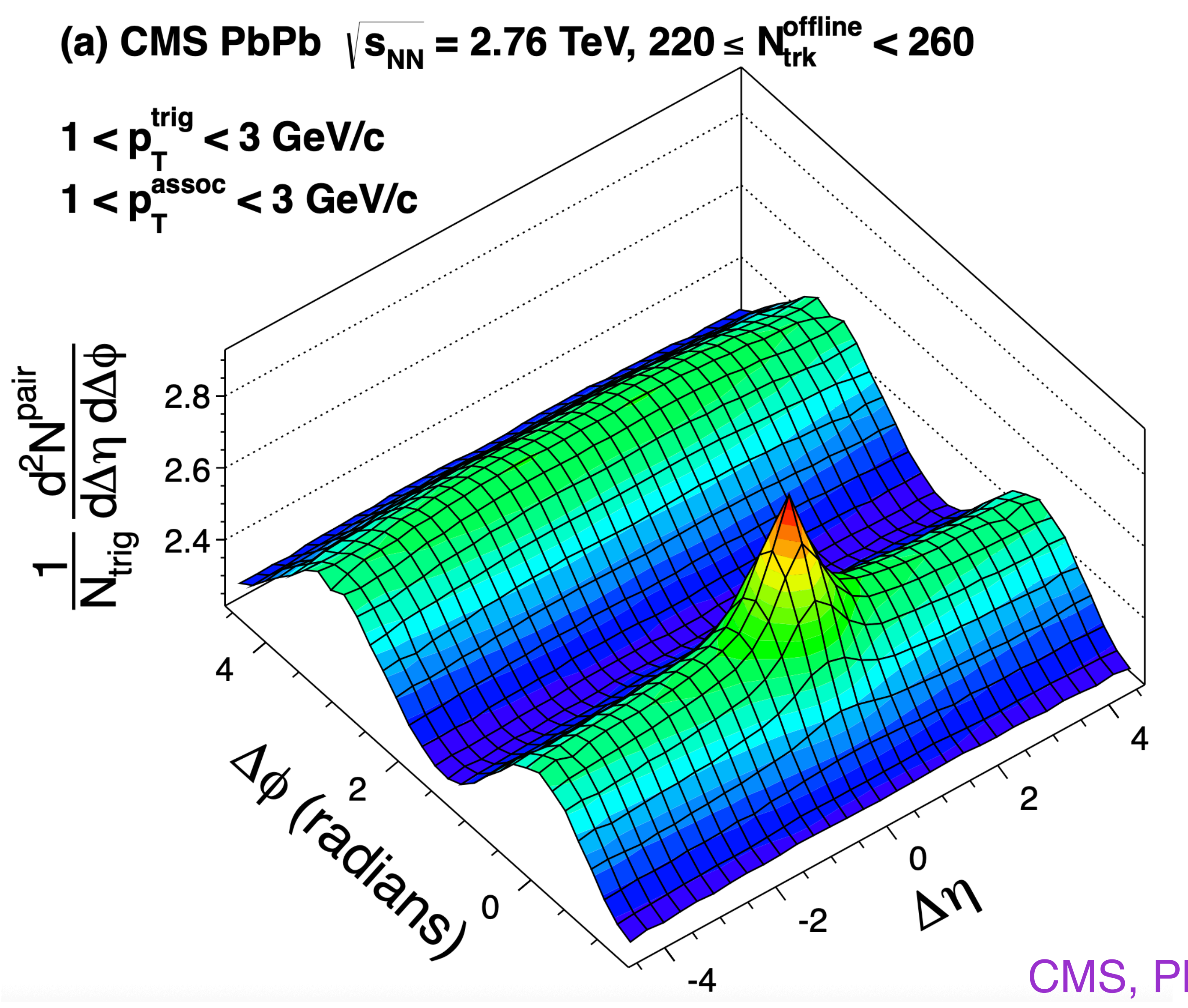
June 26, 2026

Based on arXiv: [2504.13144](), [2509.19759]()

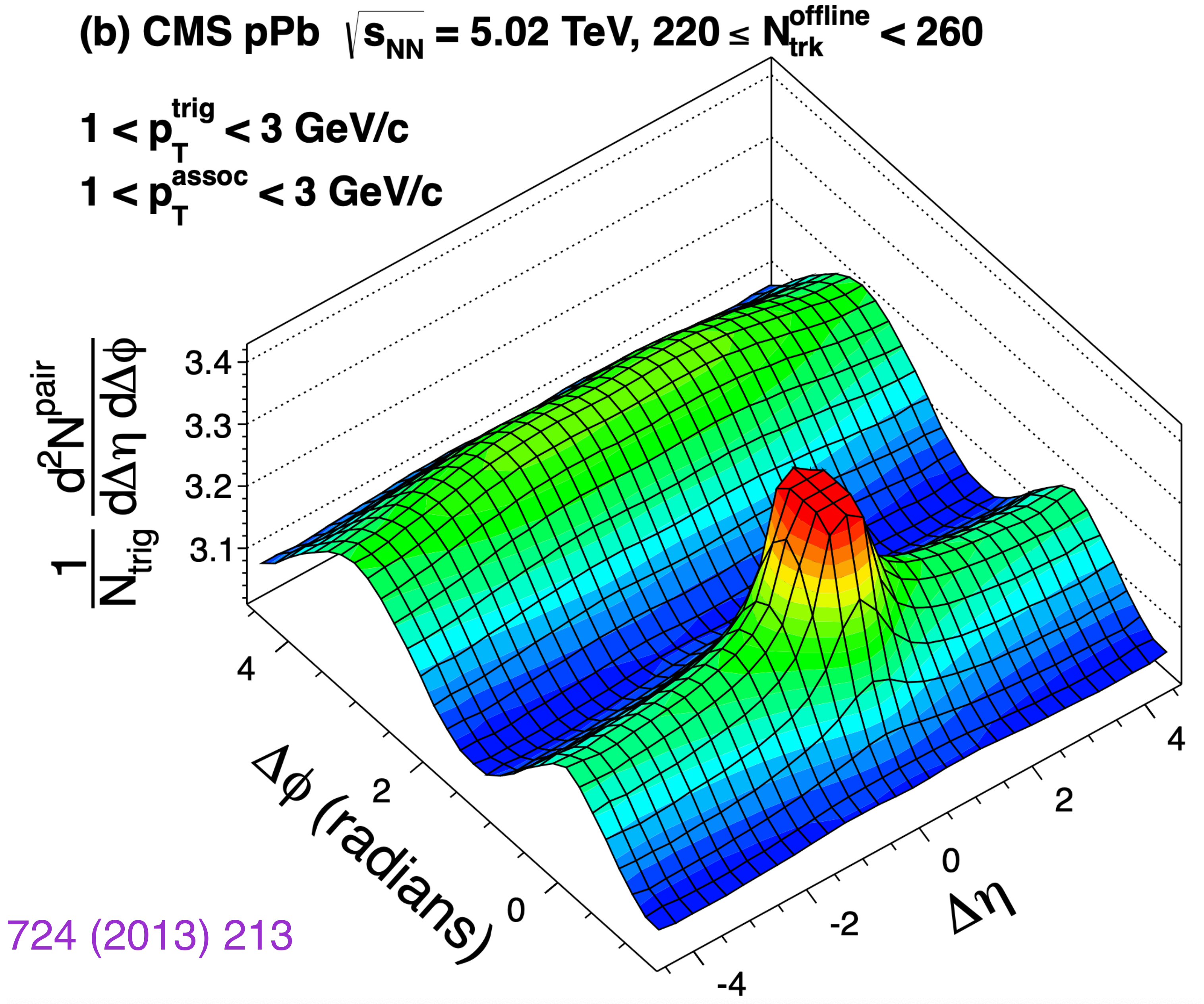
Collaborators: Richard Furnstahl, Ulrich Heinz, Matthew Pratola, Chun Shen

Everything flows?

Pb–Pb

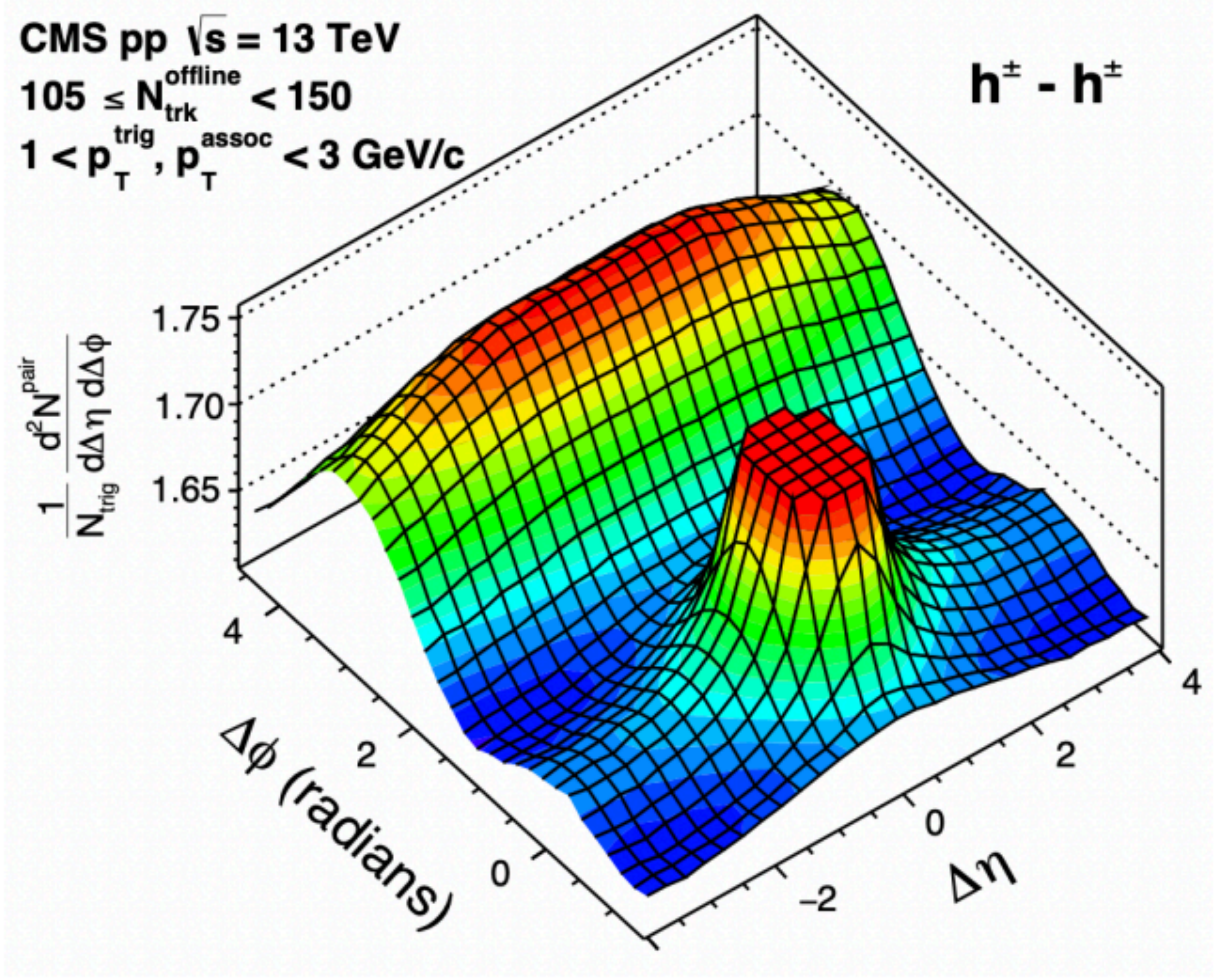


High multiplicity p–Pb



CMS, Phys. Lett. B 724 (2013) 213

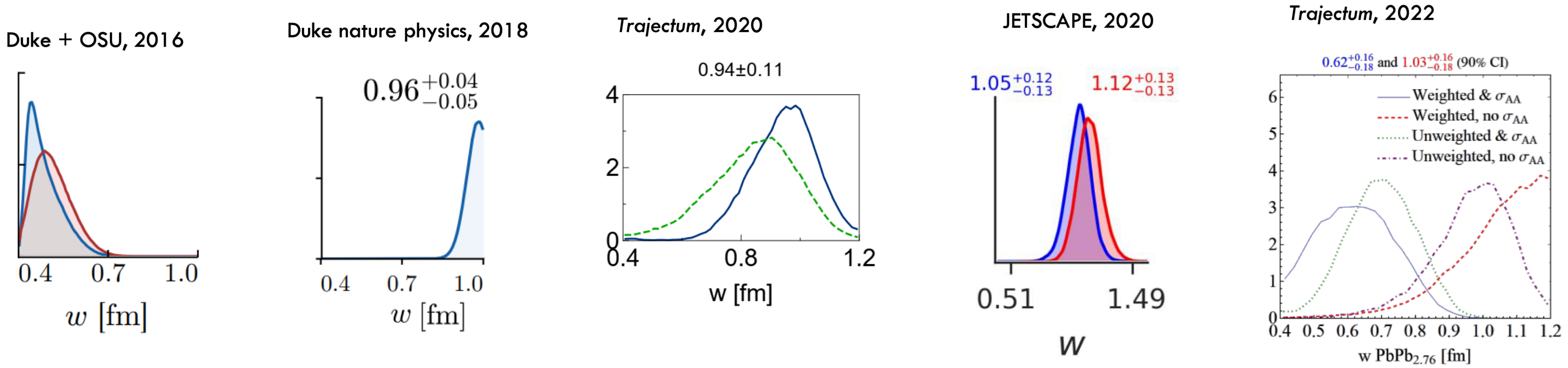
High multiplicity pp



CMS, Phys. Lett. B 765 (2017) 193

- Clear signs of long-range correlations in Pb–Pb, p–Pb, and p–p.
- Collectivity from: Hydrodynamic response, or exact conservation-induced collectivity, or initial-state effects, or…?
- We need to compare our theories with data carefully and systematically.
  Model-data comparison should aid our understanding in identifying inadequacies of our theoretical models.

Model-to-data comparison : Caution



Nucleon width increased in 2018 (very significantly) and decreased (very significantly) in 2022

- Turns out that the width was inconsistent with the total hadronic PbPb cross section (measured in 2022)

What happened?

- In 2018: the initial stage model changed
- In 2022: new measurement, but also:
  Many subtle changes in many data points `pulled' to large width

Wilke van der Schee
Student day Quark Matter 2025
6 April 2025

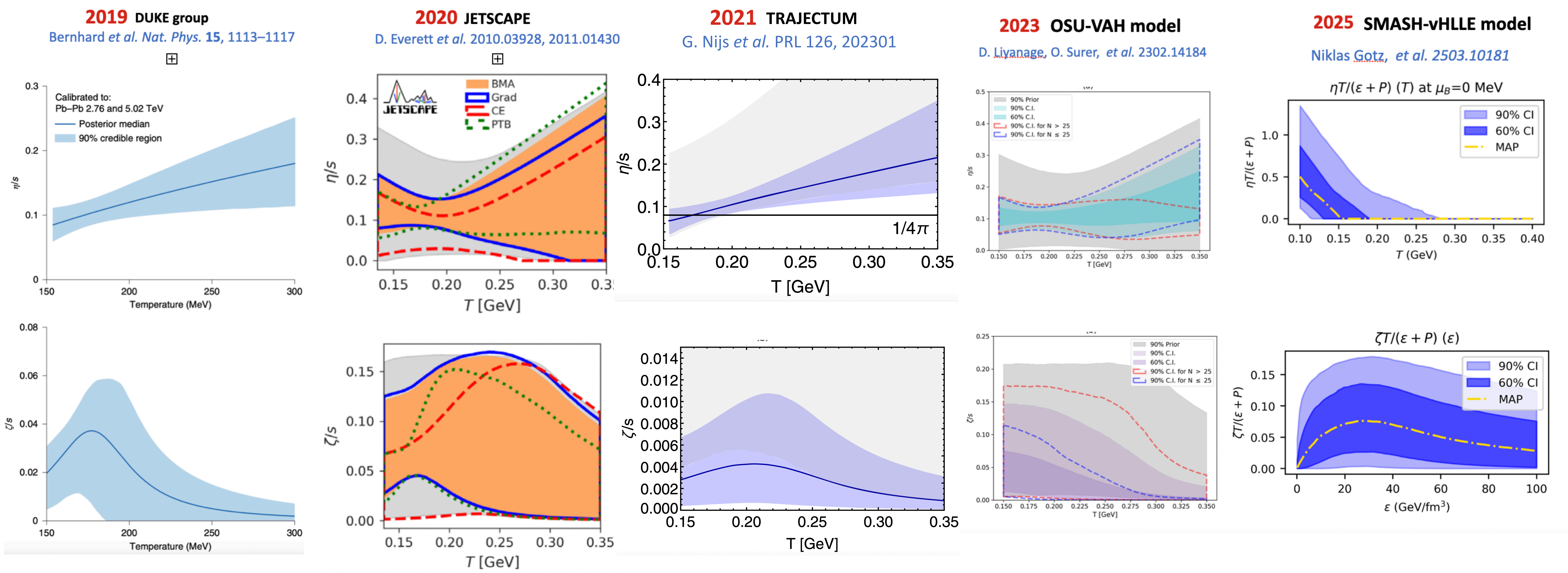
Giuliano Giacalone, arXiv:2208.06839

- Change in physics models and/or data **changes physical parameters**.
- **Experimental uncertainties are accounted** for in Bayesian model-data comparison.
- **Theoretical uncertainties are not quantified and accounted** → Approximate models forced to fit data outside their domain of validities → parameters become fitting variables.

Changing nucleon width and specific viscosities with change in model/data

η/s →

ζ/s →



A simple example: Ball drop experiment

[SJ, Shen, Furnstahl, Heinz, Pratola, PLB 870, 139946 (2025)]

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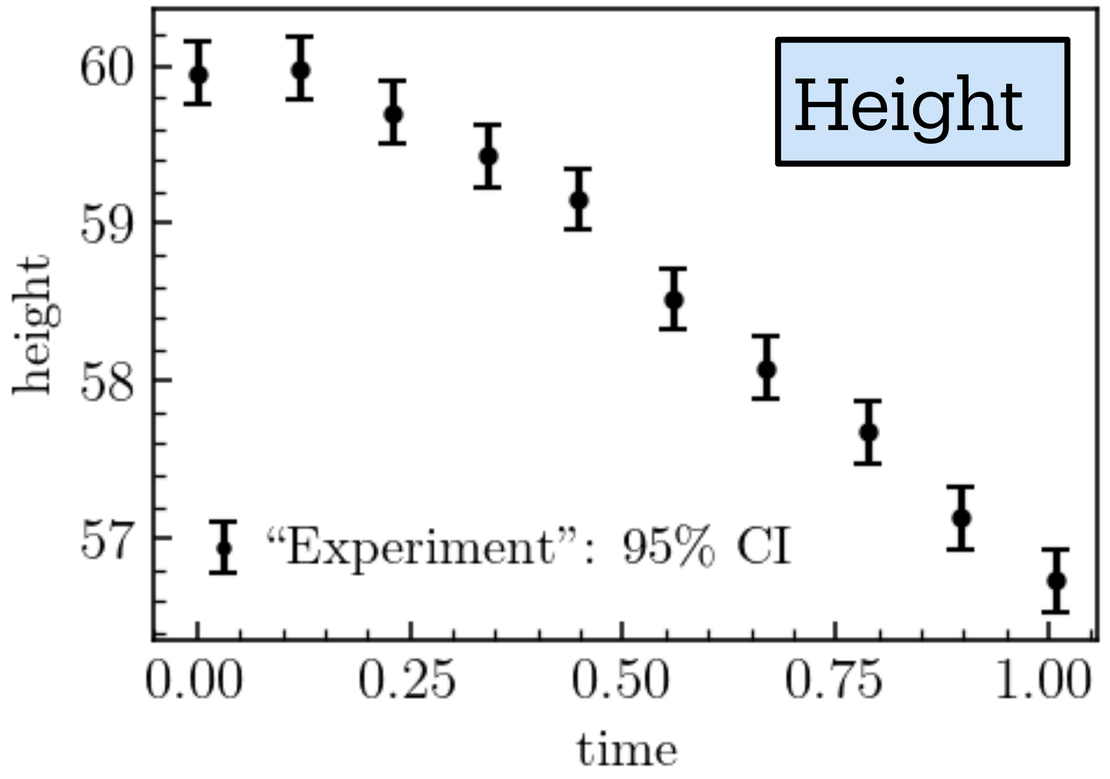

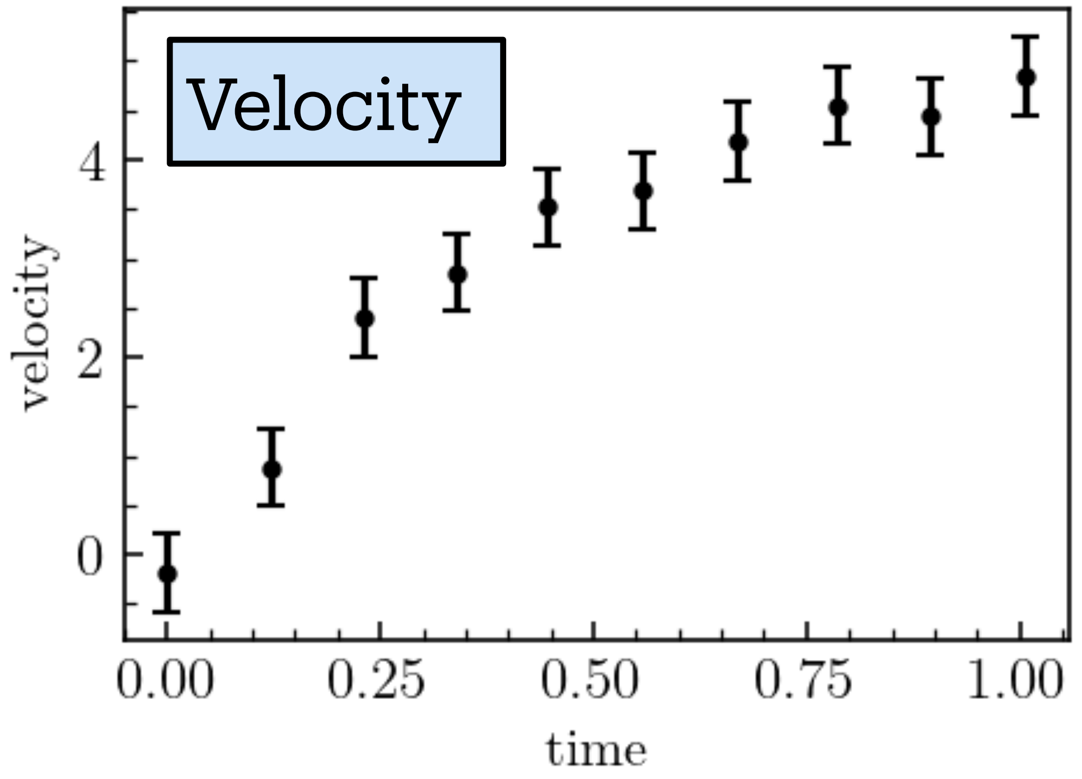

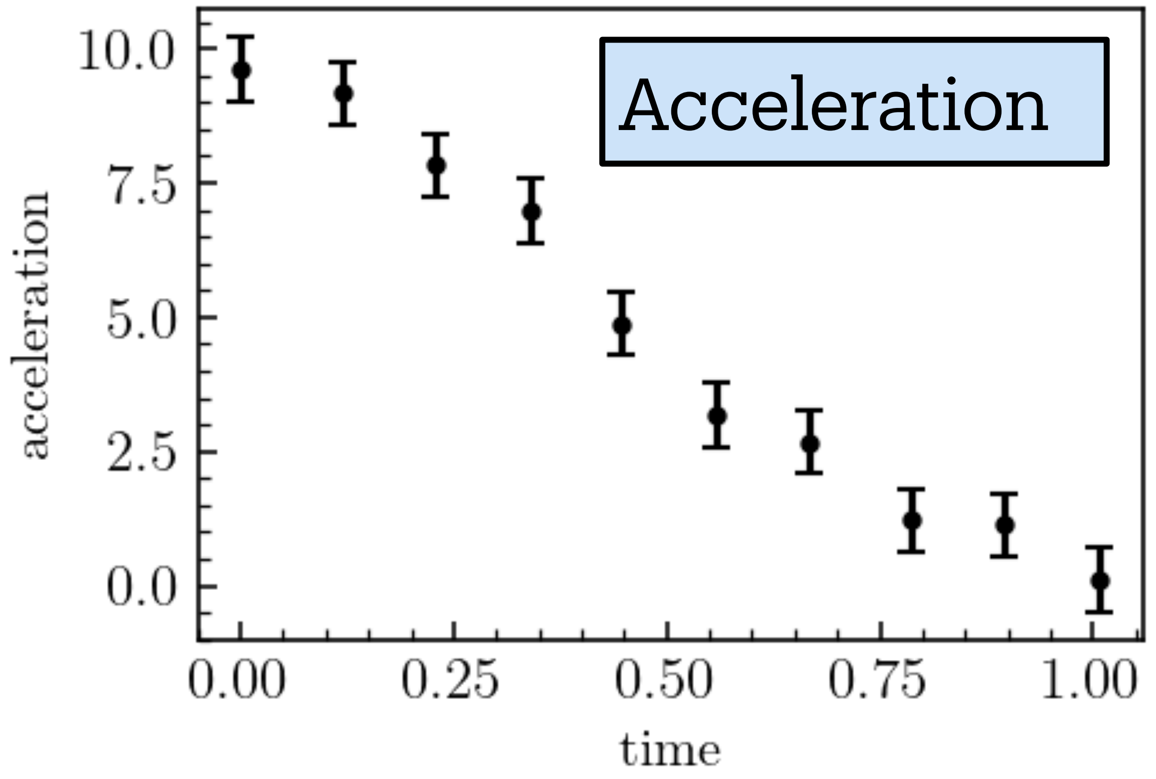

Drag force: $\mathbf{f}_D = -0.4v^2\,\hat{\mathbf{v}}$

EoM:

$$a = \frac{d\mathbf{v}}{dt} = \mathbf{g} - 0.4v^2\hat{\mathbf{v}}\,, \quad \frac{dh}{dt} = -v$$

$h_0 = 60\,\mathrm{m}, v_0 = 0\,\mathrm{m/s}, g = 9.8\,\mathrm{m/s^2}$

Theoretical model (theory): EoM: $h = h_0 - v_0 t - \frac{1}{2}gt^2, \quad v = v_0 + gt, \quad a = g$

2 model parameters : $\theta = (g, v_0)$

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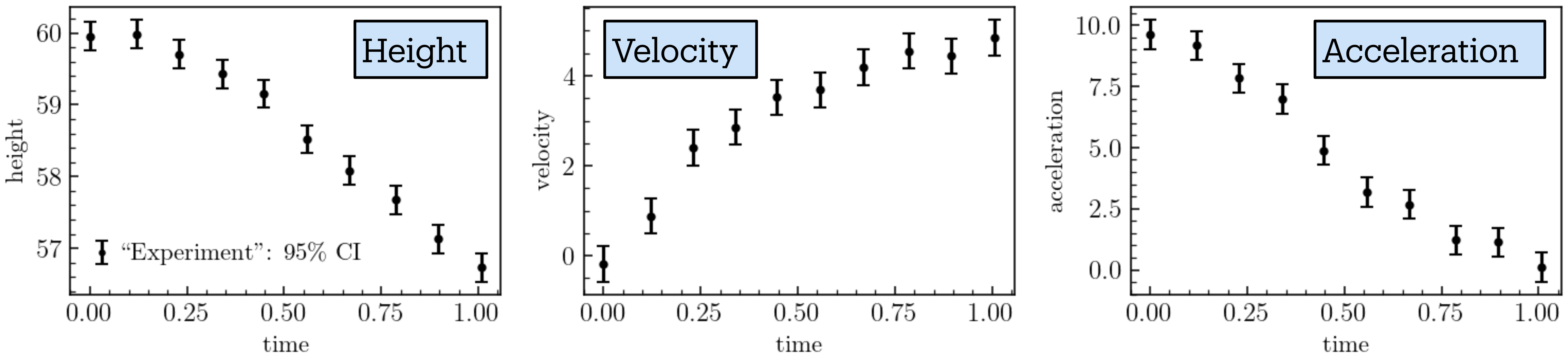

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Bayesian updating of knowledge

$$\mathrm{pr}(A|B,I) = \frac{\mathrm{pr}(B|A,I)\,\mathrm{pr}(A|I)}{\mathrm{pr}(B|I)} \Longrightarrow \underbrace{\mathrm{pr}(\alpha|\mathbf{y}_{\mathrm{exp}}, I)}_{\text{posterior}} \propto \underbrace{\mathrm{pr}(\mathbf{y}_{\mathrm{exp}}|\alpha, I)}_{\text{likelihood}} \times \underbrace{\mathrm{pr}(\alpha|I)}_{\text{prior}}$$

$$\underbrace{y_{\mathrm{exp}}(t)}_{\text{Experimental observation}} = \underbrace{\eta_{\mathrm{mod}}(t,\theta)}_{\text{Predictions from approximate theory}} + \underbrace{\delta_{\mathrm{MD}}(t)}_{\text{Error of approximate theory predictions}} + \underbrace{\epsilon(t)}_{\text{Observation error}}$$

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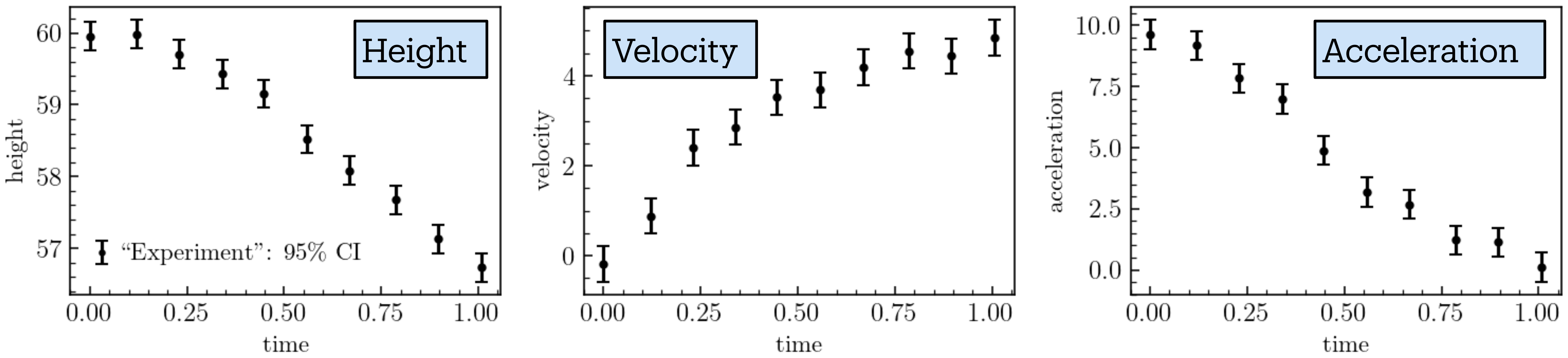

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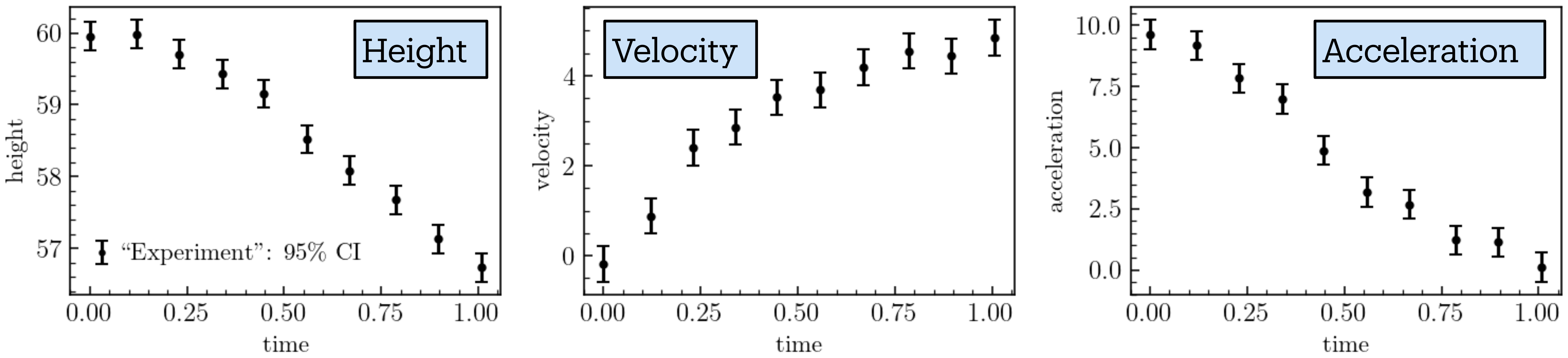

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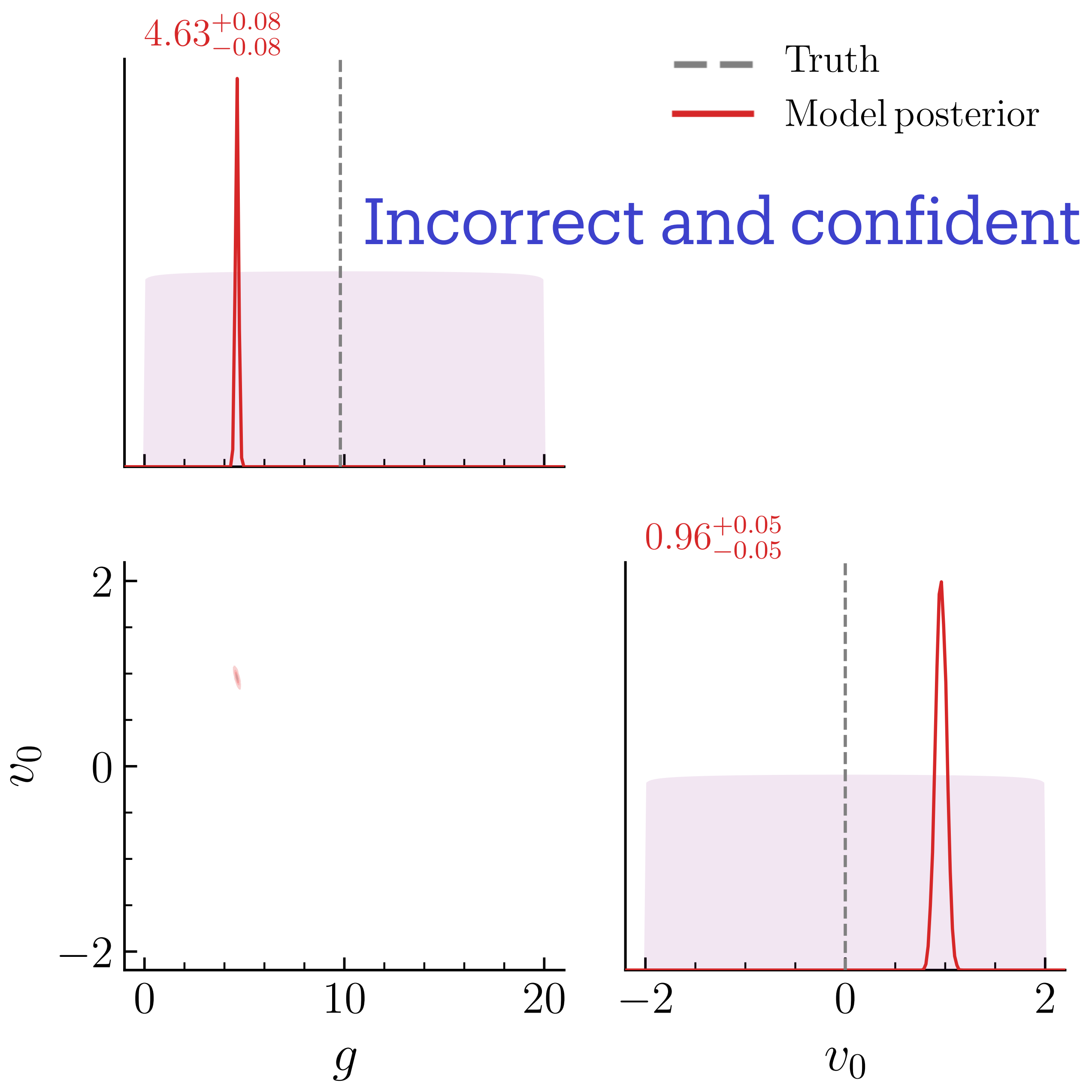
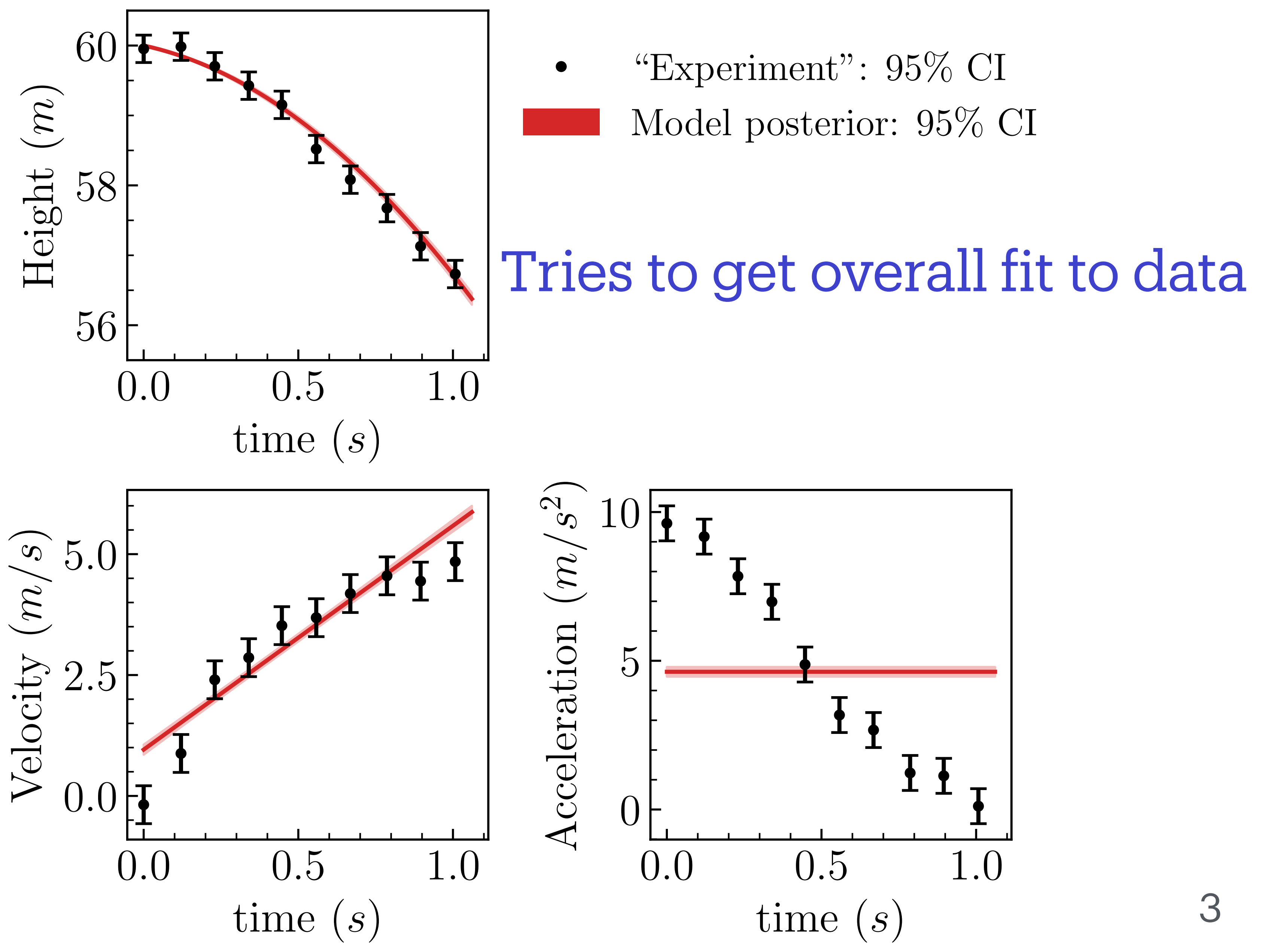




Model discrepancy framework

$$\underbrace{y_{\mathrm{exp}}(t)}_{\substack{\text{Experimental}\\ \text{observation}}} = \underbrace{\eta_{\mathrm{mod}}(t,\theta)}_{\substack{\text{Predictions from}\\ \text{approximate theory}}} + \underbrace{\delta_{\mathrm{MD}}(t)}_{\substack{\text{Error of approximate}\\ \text{theory predictions}}} + \underbrace{\epsilon(t)}_{\substack{\text{Observation}\\ \text{error}}}$$

M. Kennedy, A. O'Hagan
doi:10.1111/1467-9868.00294

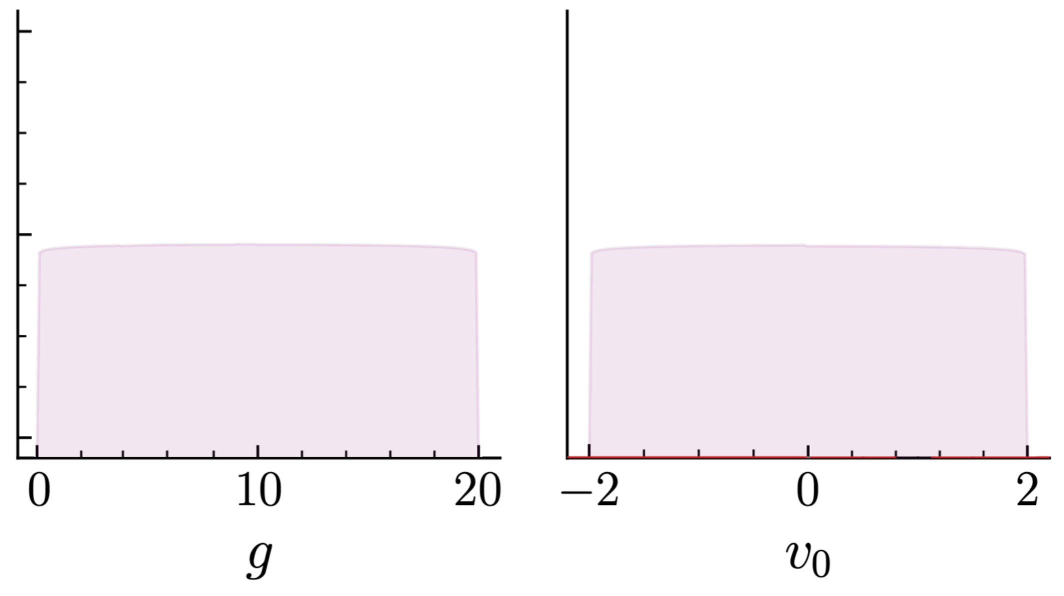
J. Brynjarsdóttir and A. O'Hagan,
doi:10.1088/0266-5611/30/11/114007

D. Higdon, M. Kennedy, *et. al.*,
doi:10.1137/S1064827503426693

- Infer **model parameters θ together with $\delta_{\mathrm{MD}}(t)$ from data.**

[SJ, Shen, Furnstahl, Heinz, Pratola, PLB 870, 139946 (2025)]

- Priors for parameters θ: g **non-negative and initial velocity small.**



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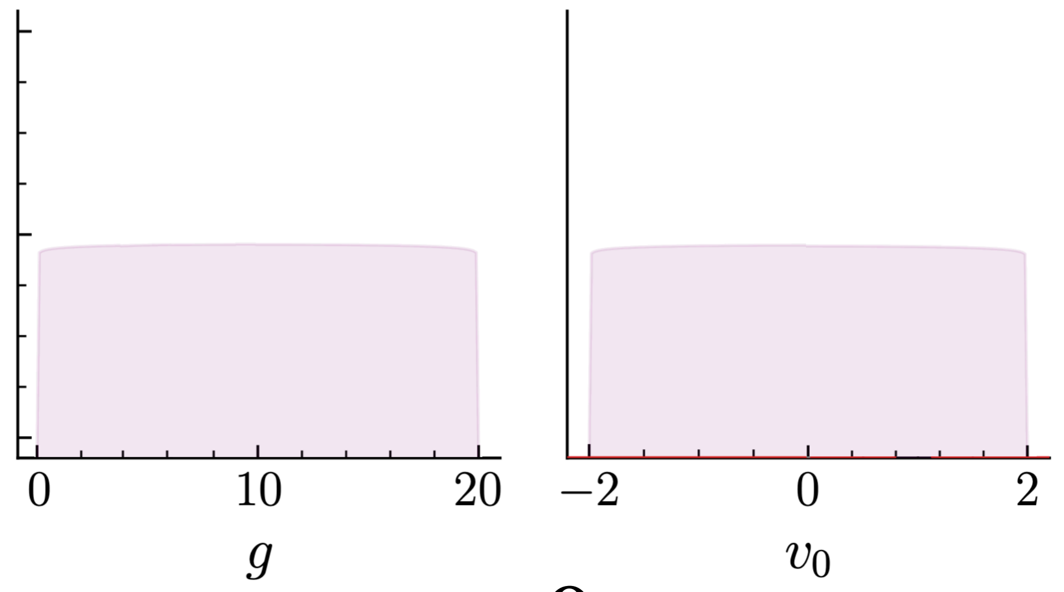

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  => **Theory more correct at small t than at large t and correct at $t = 0$ (initial velocity near 0).**
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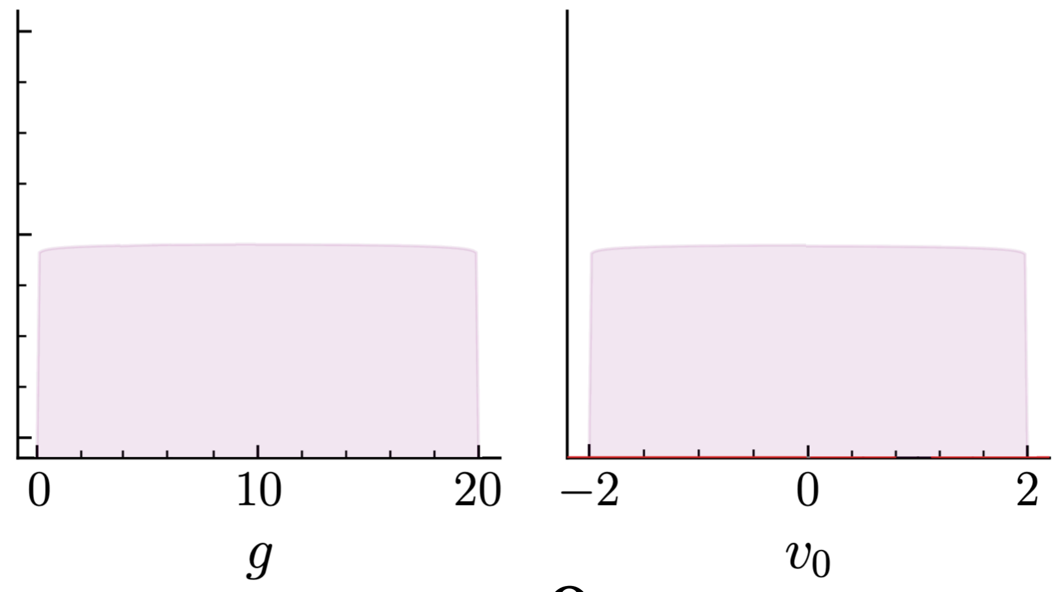

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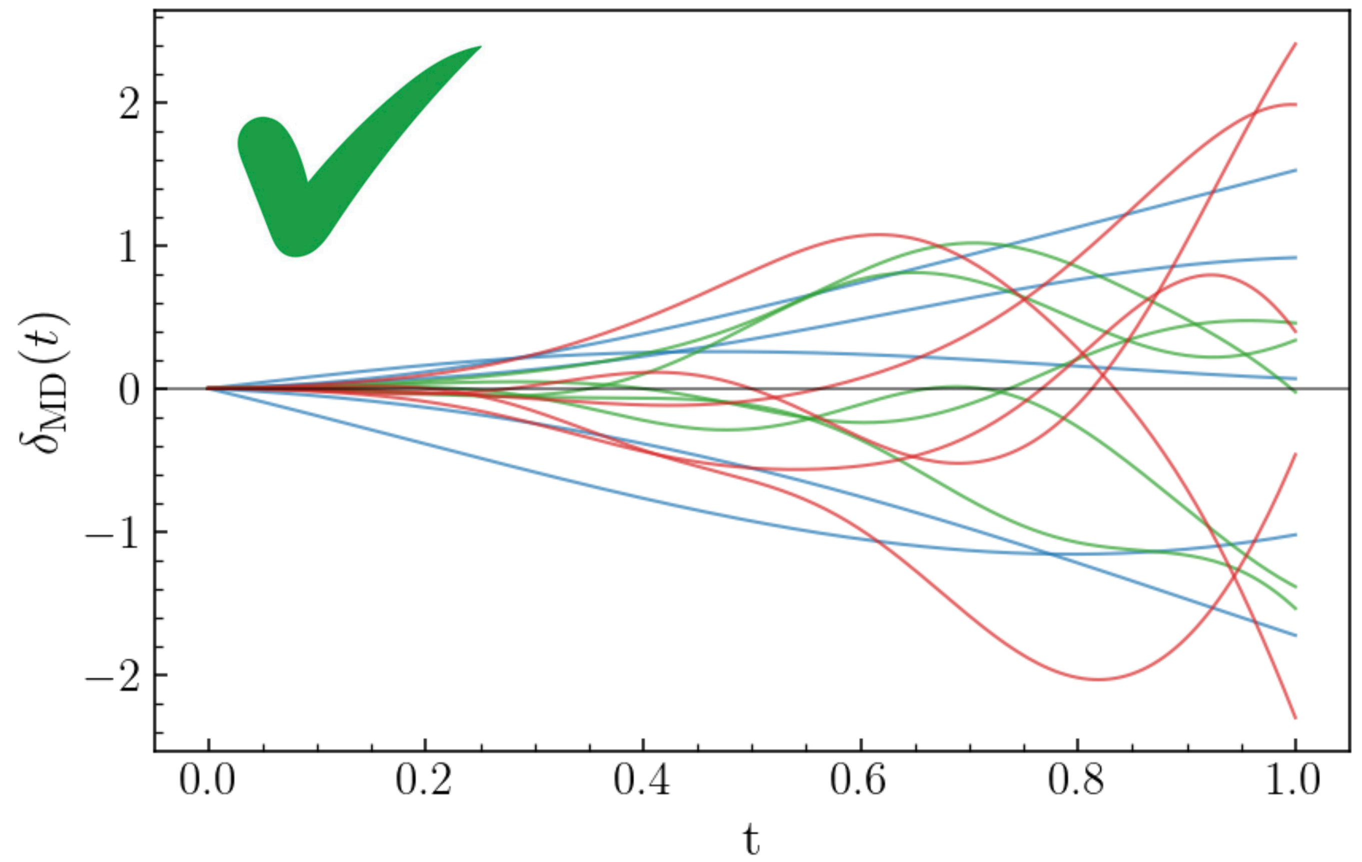

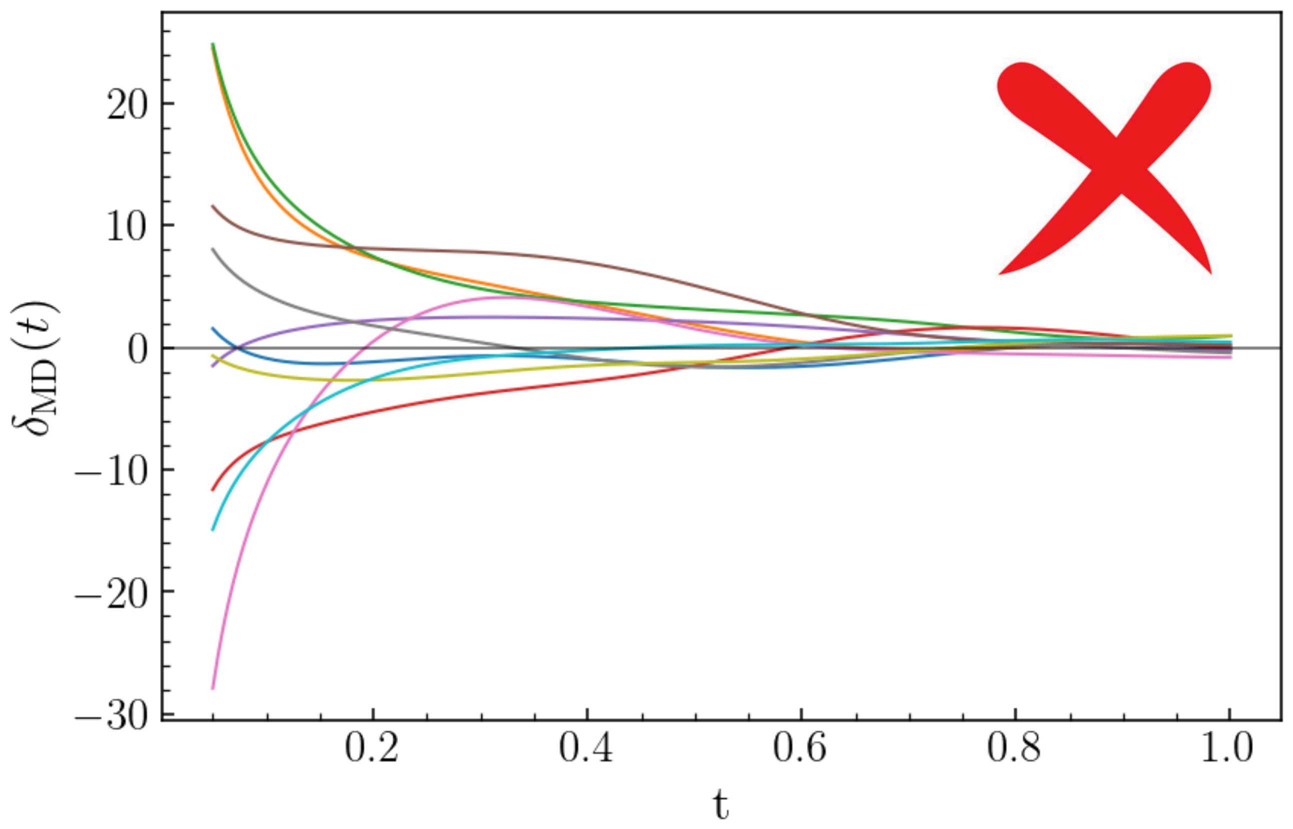

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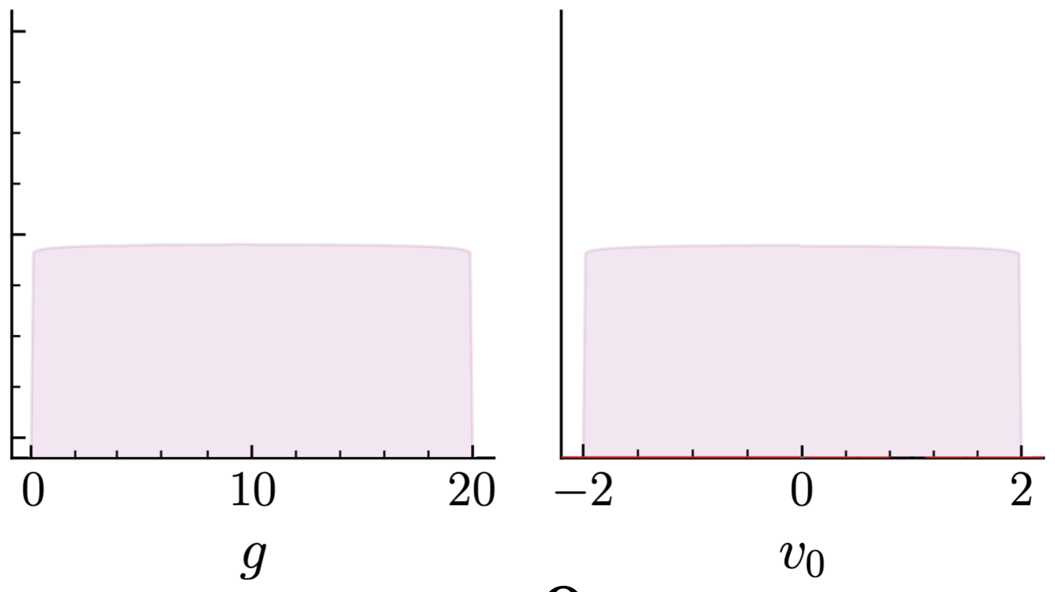

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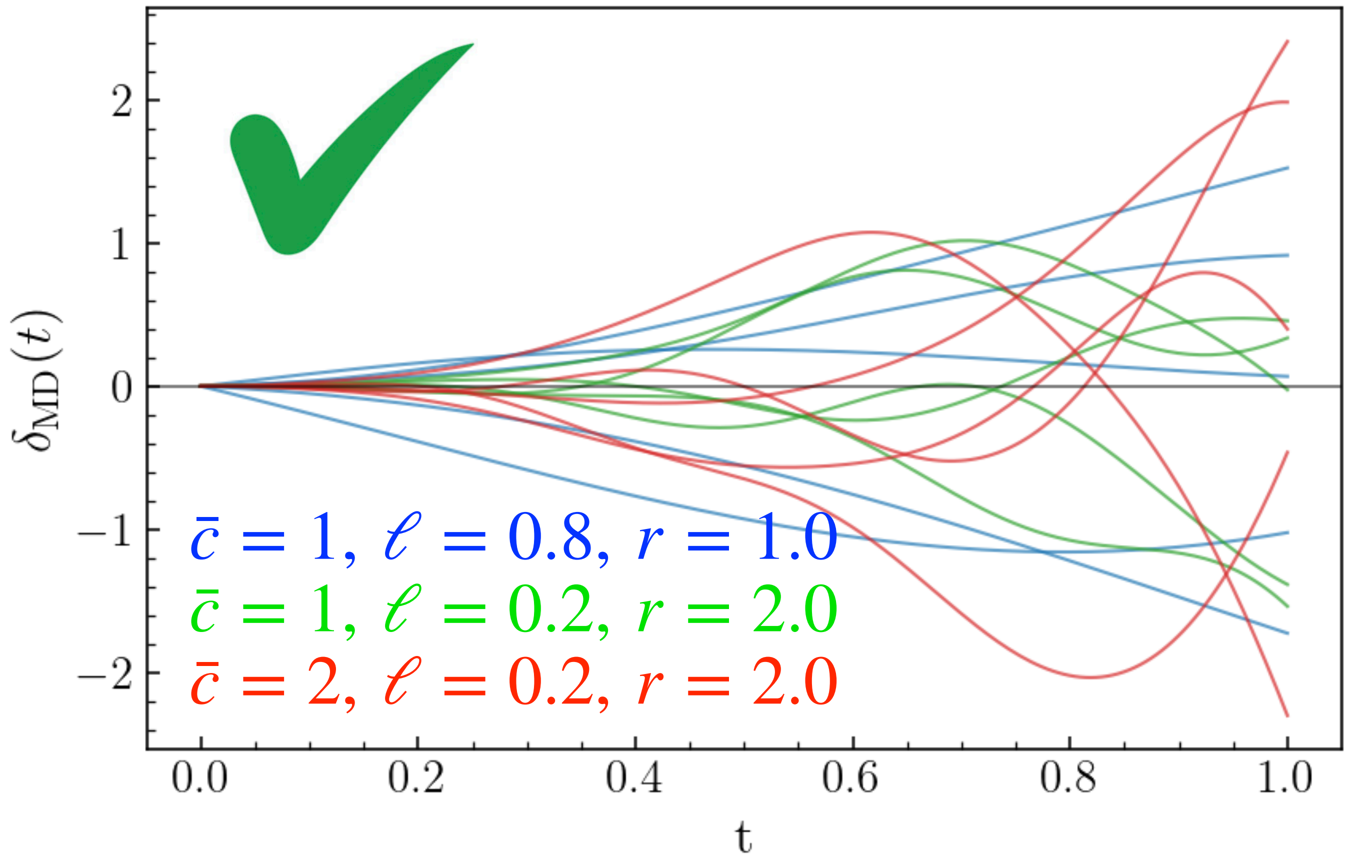
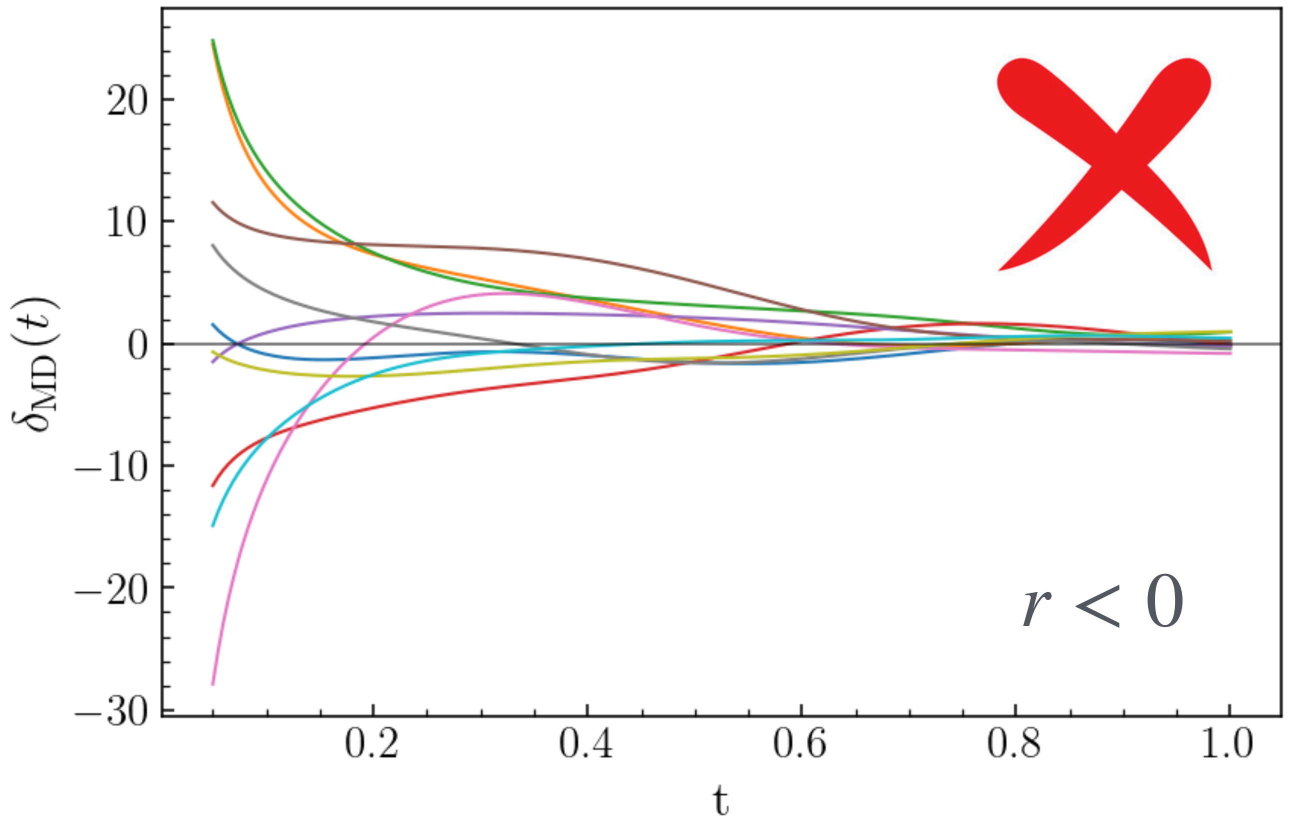
$$\delta(\,\cdot\mid\phi) \sim \text{GP}\left(\mathbf{0}, K(\,\cdot\,,\,\cdot\mid\phi)\right).$$

A Gaussian process is a probability distribution over functions.

$$K(t_i, t_j \mid \phi) \equiv \bar{c}^2 (t_i t_j)^r \exp\left(-\frac{\|t_i - t_j\|^2}{2\ell^2}\right). \qquad r \geq 0$$





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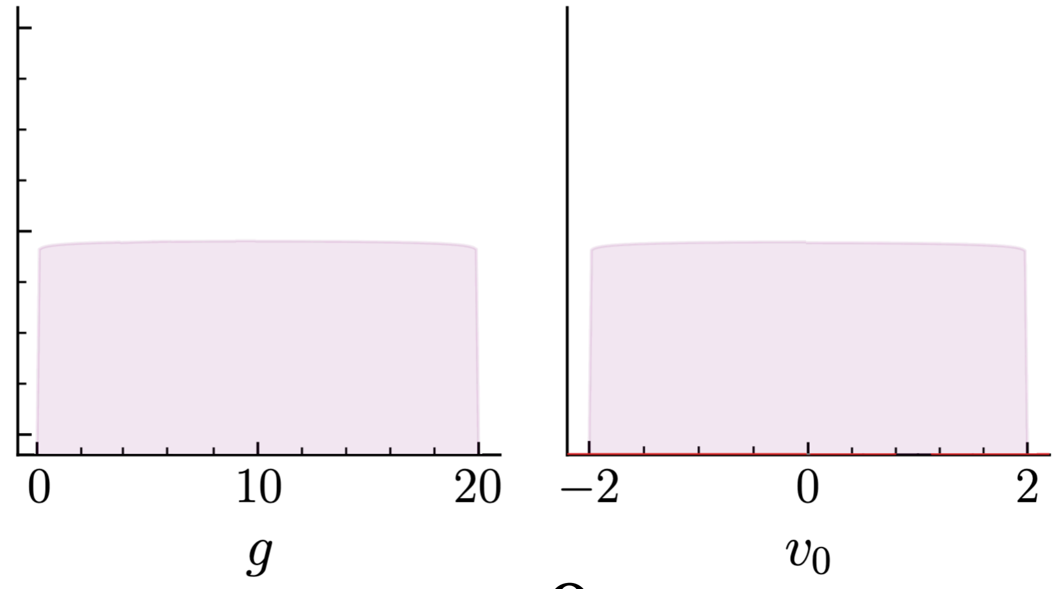

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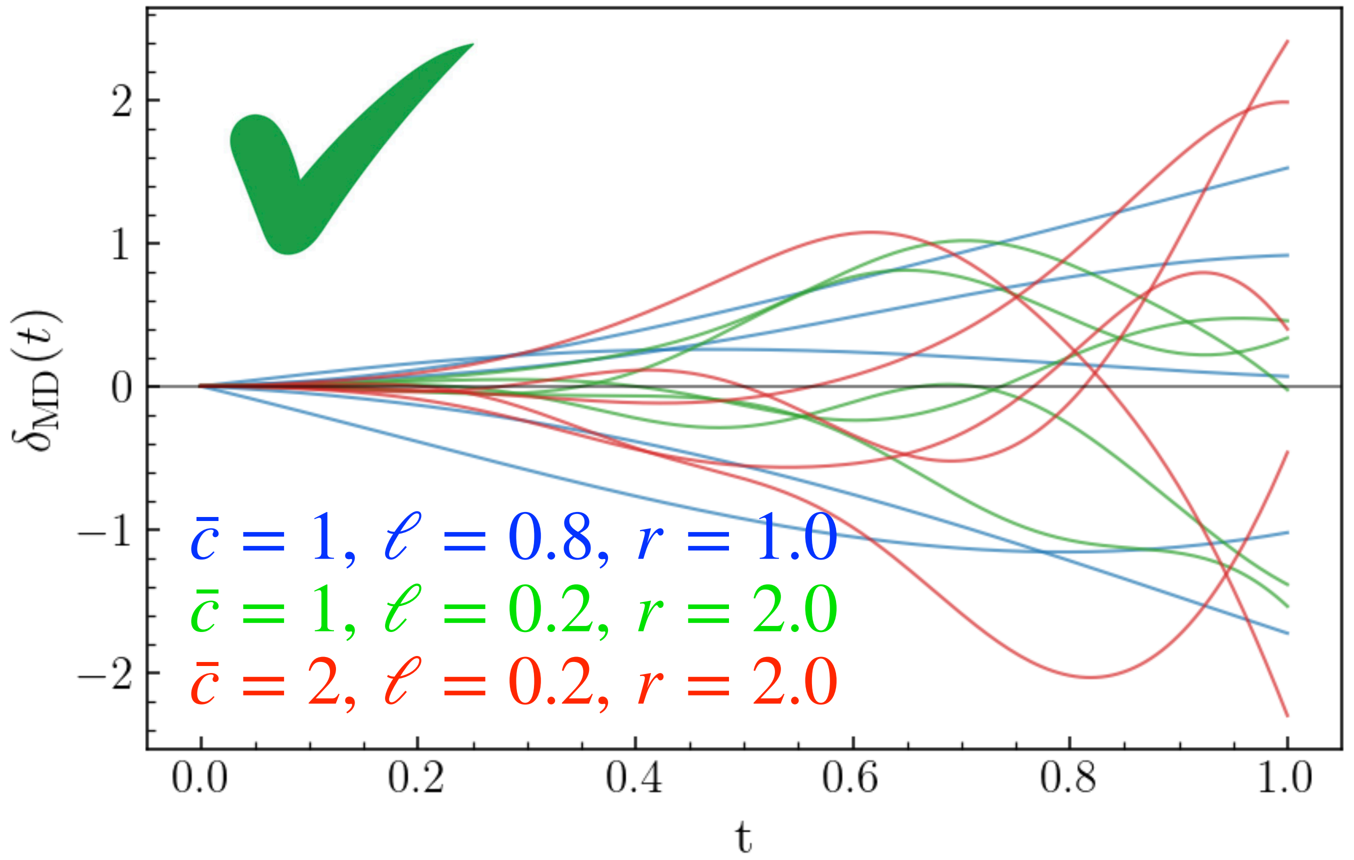
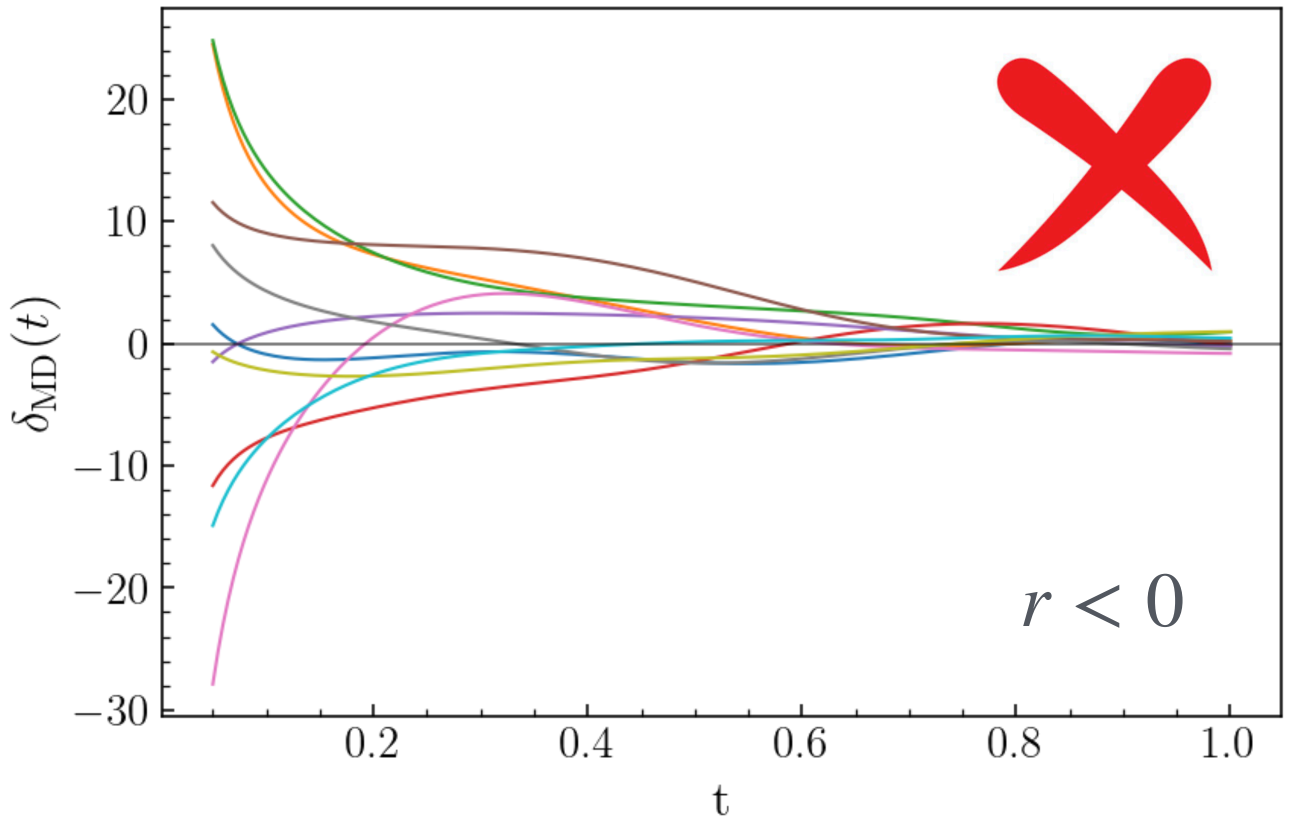
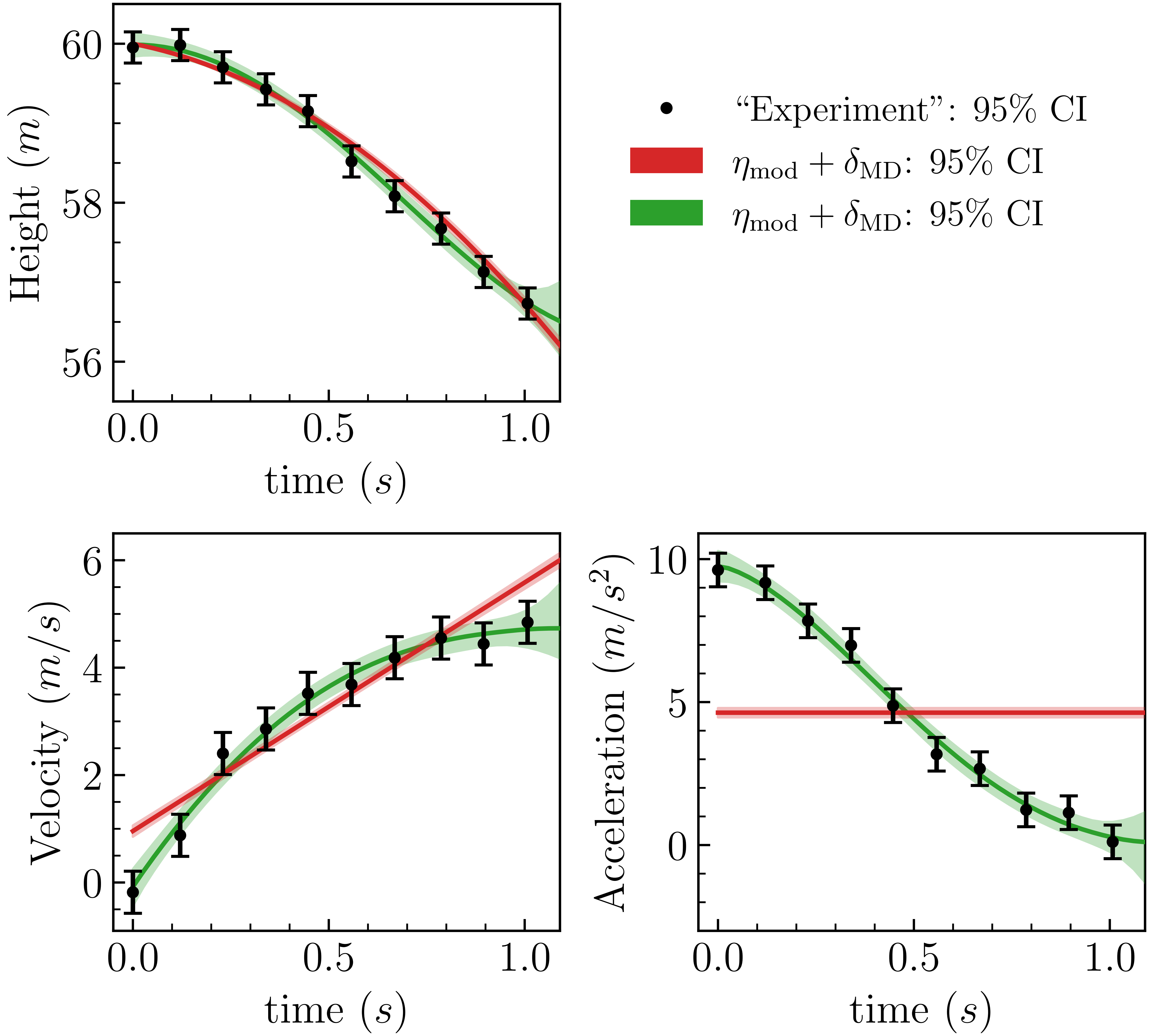




Fit model parameters θ and GP hyperparameters ϕ simultaneously to data.

Ball drop experiment with model discrepancy

[SJ, Shen, Furnstahl, Heinz, Pratola, PLB 870, 139946 (2025)]



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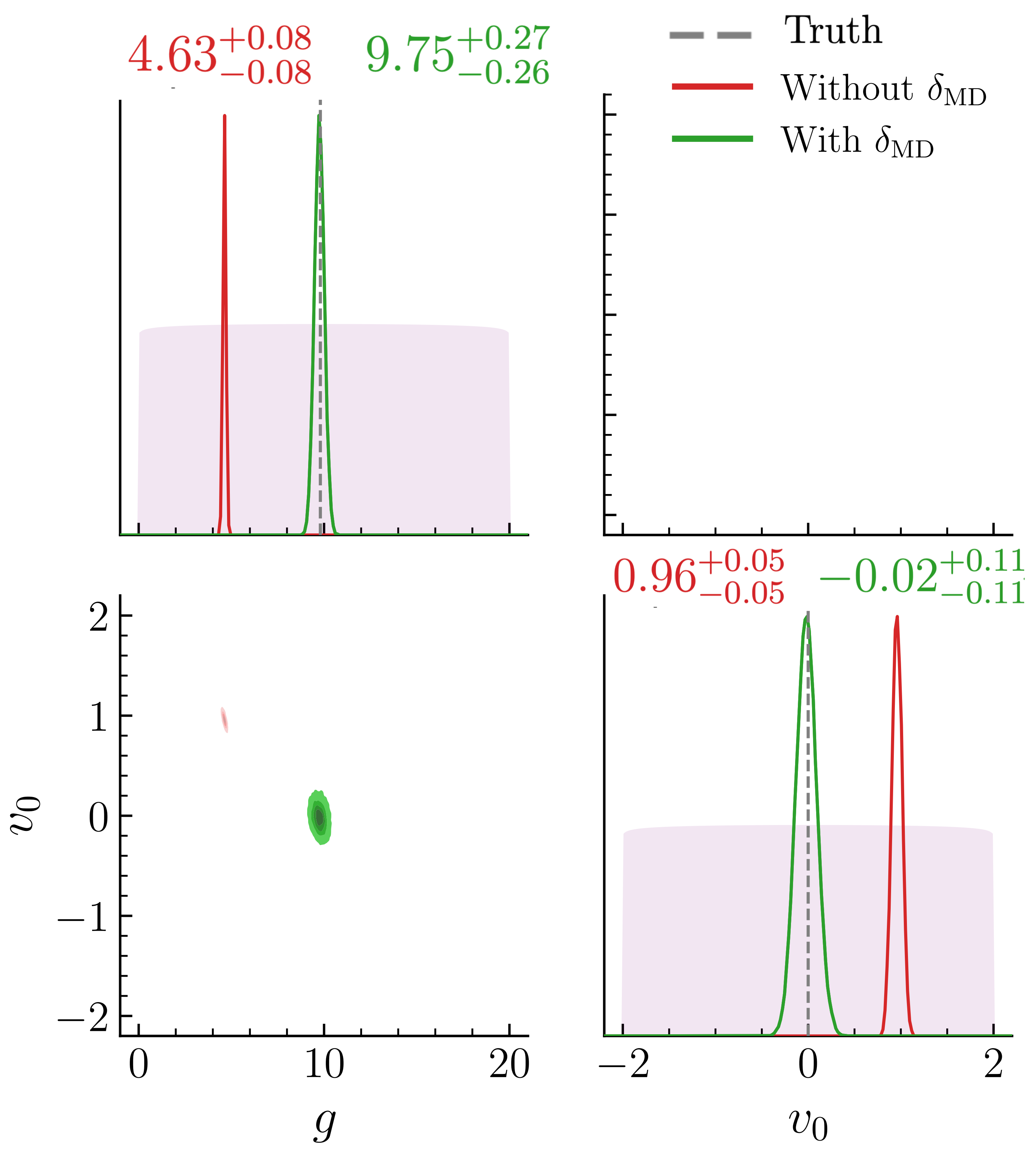

Correct and confident!

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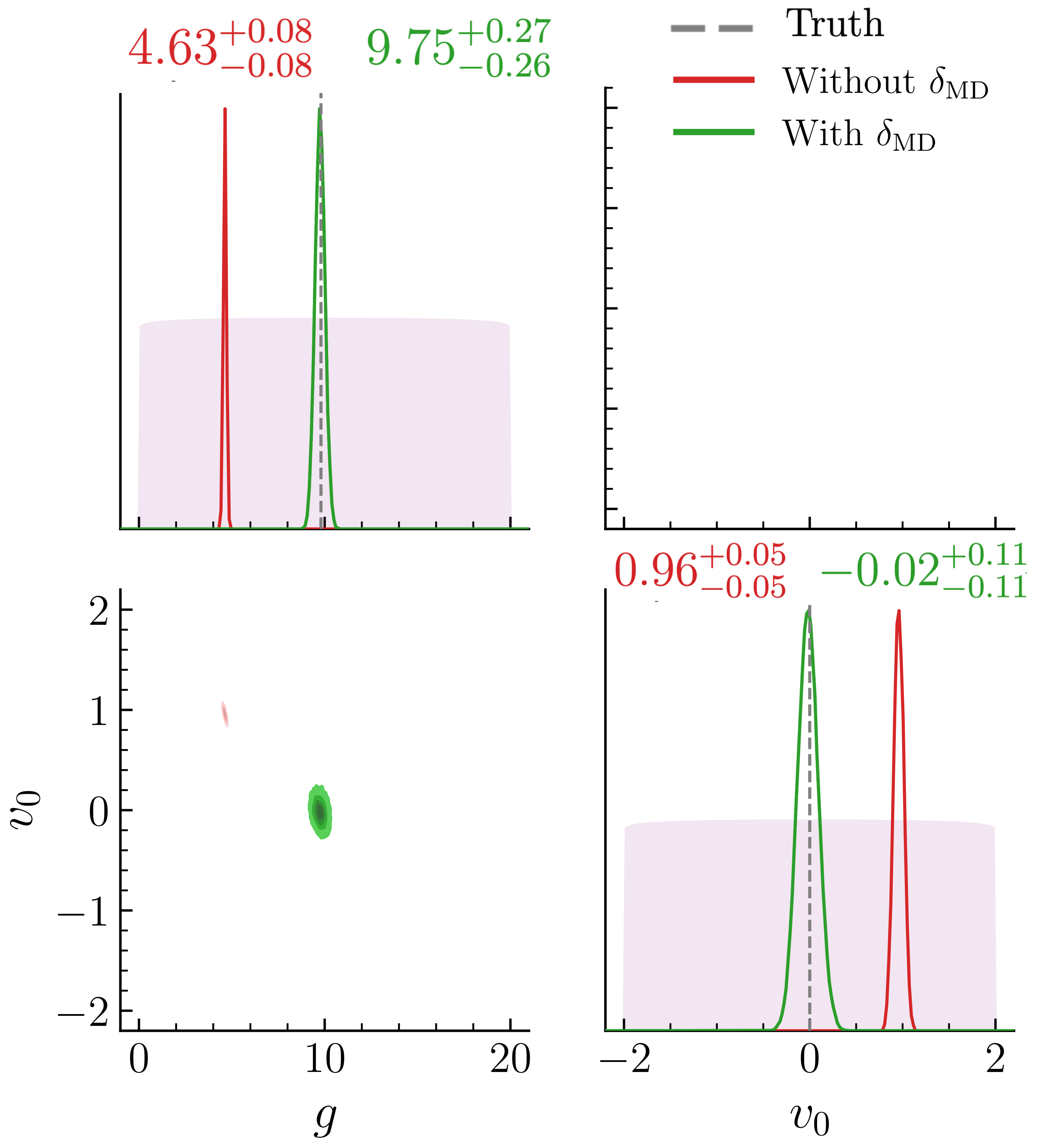

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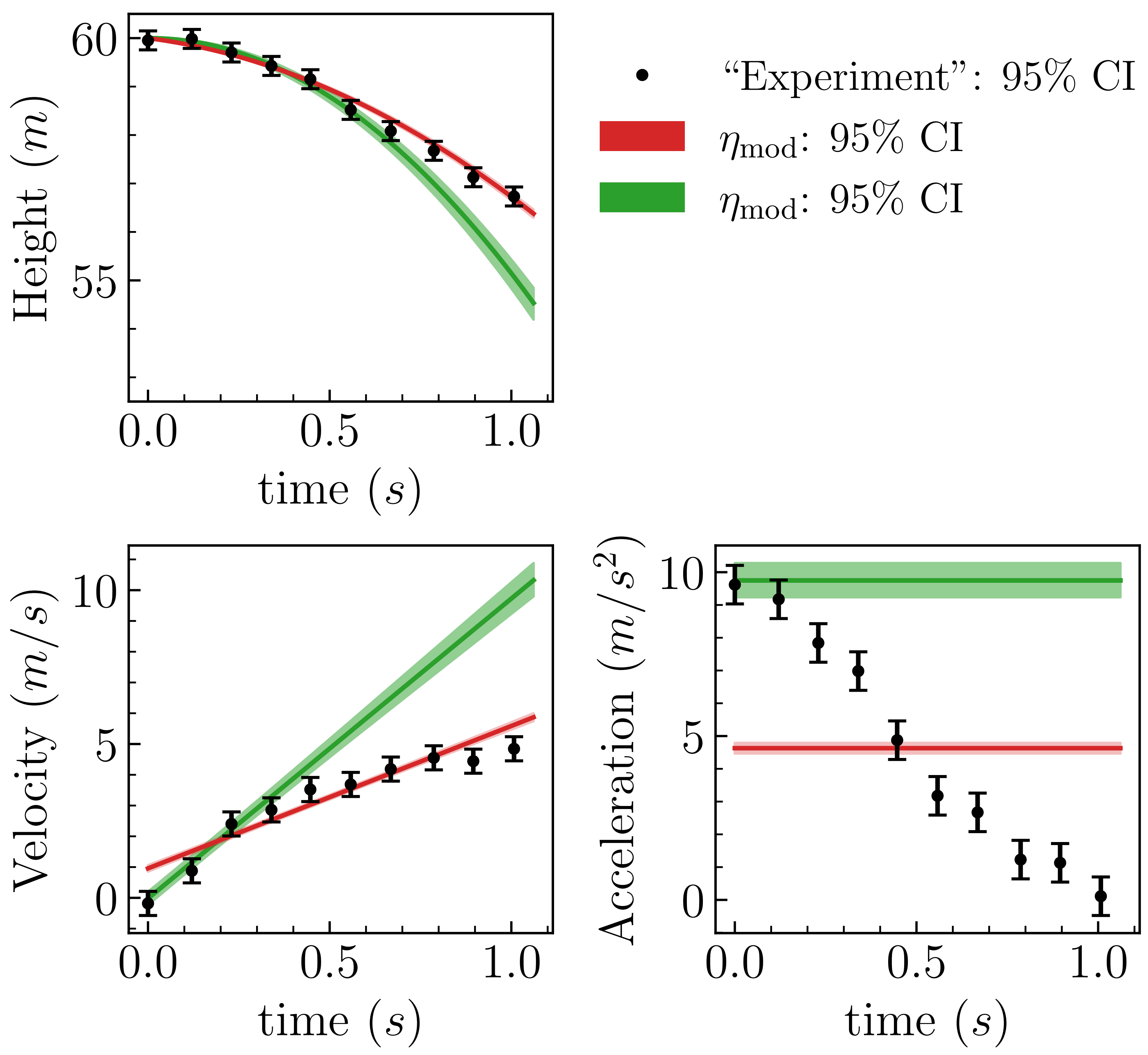

Fits data better around $t = 0$ where the model is correct.

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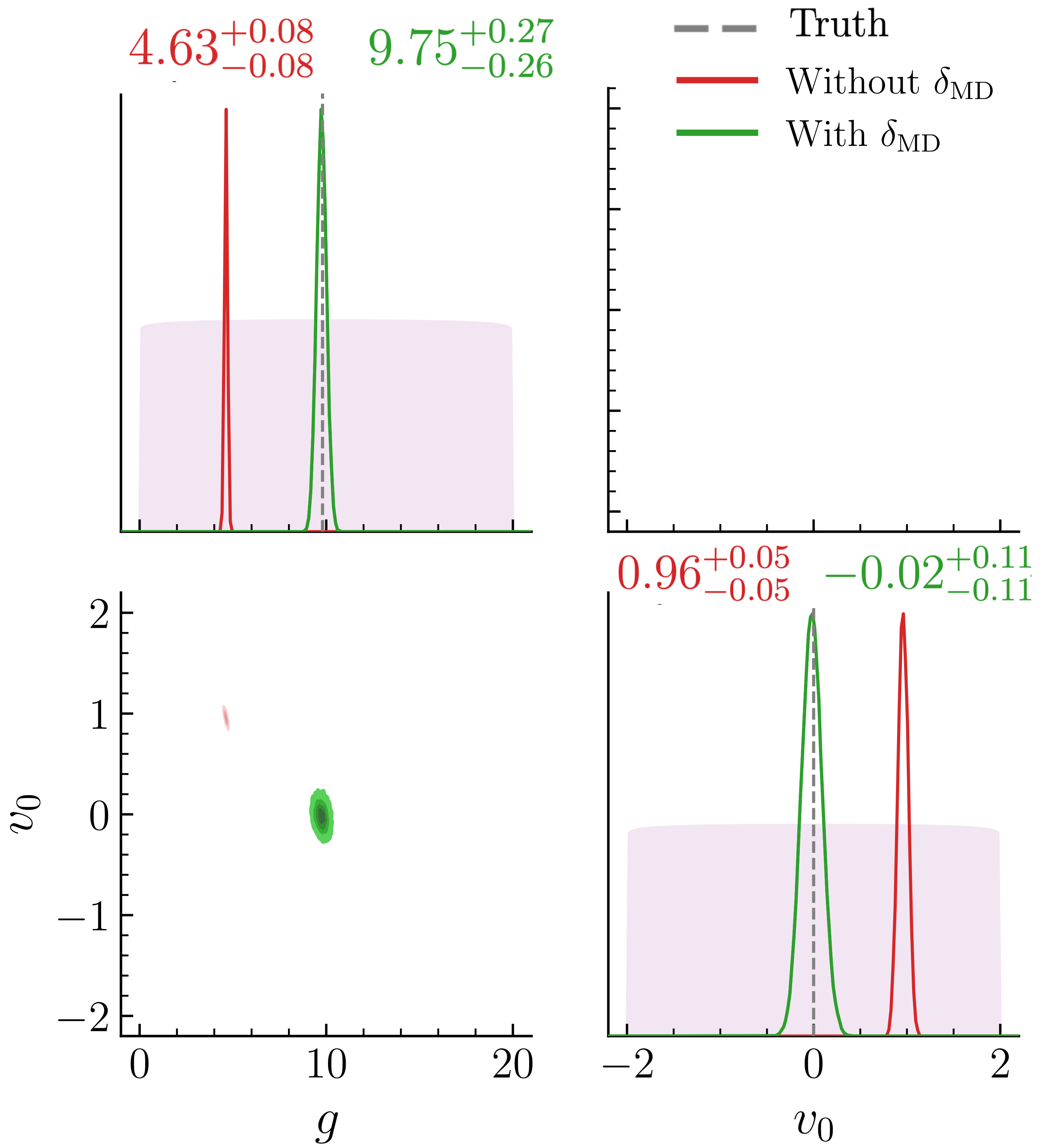

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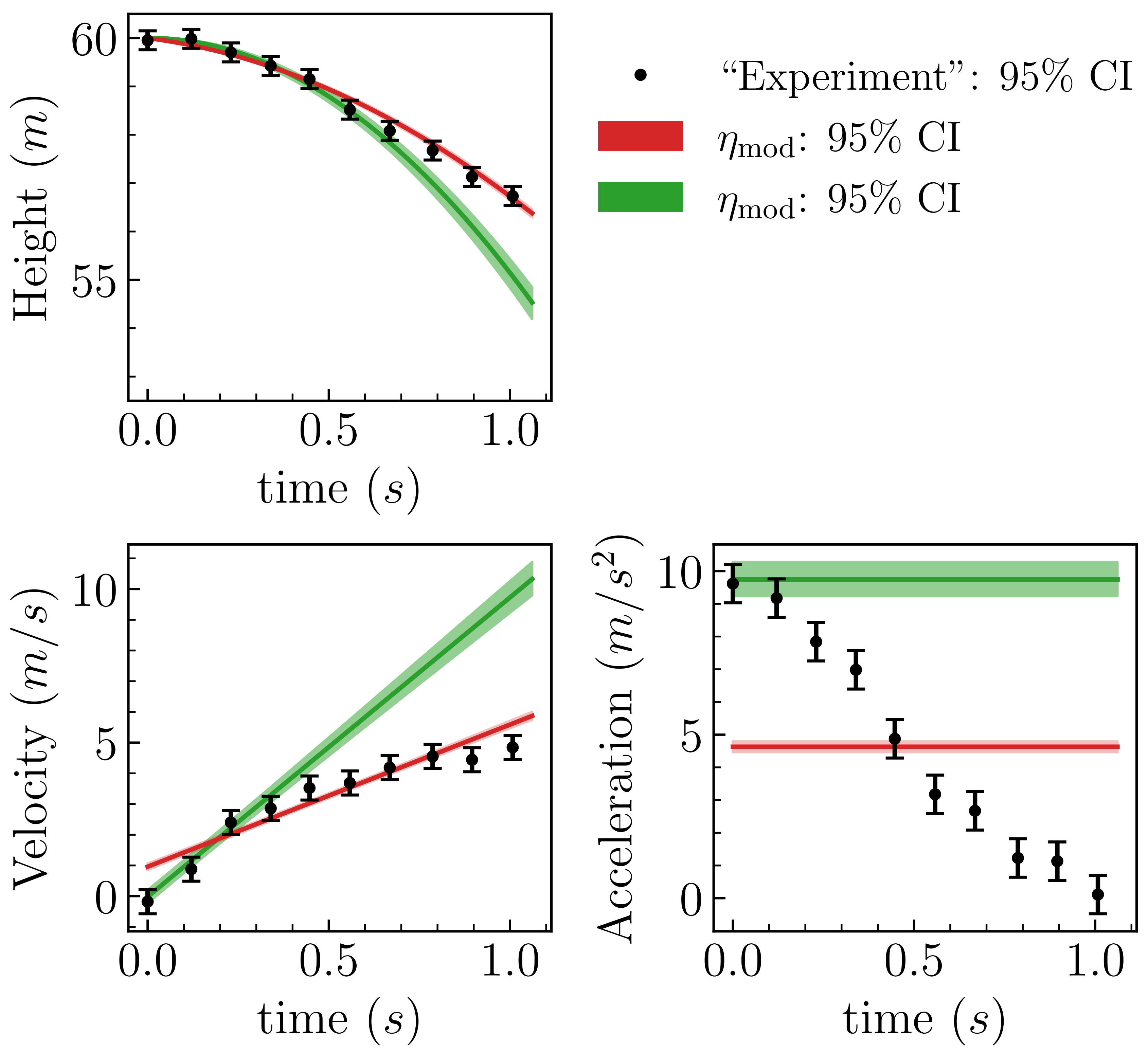

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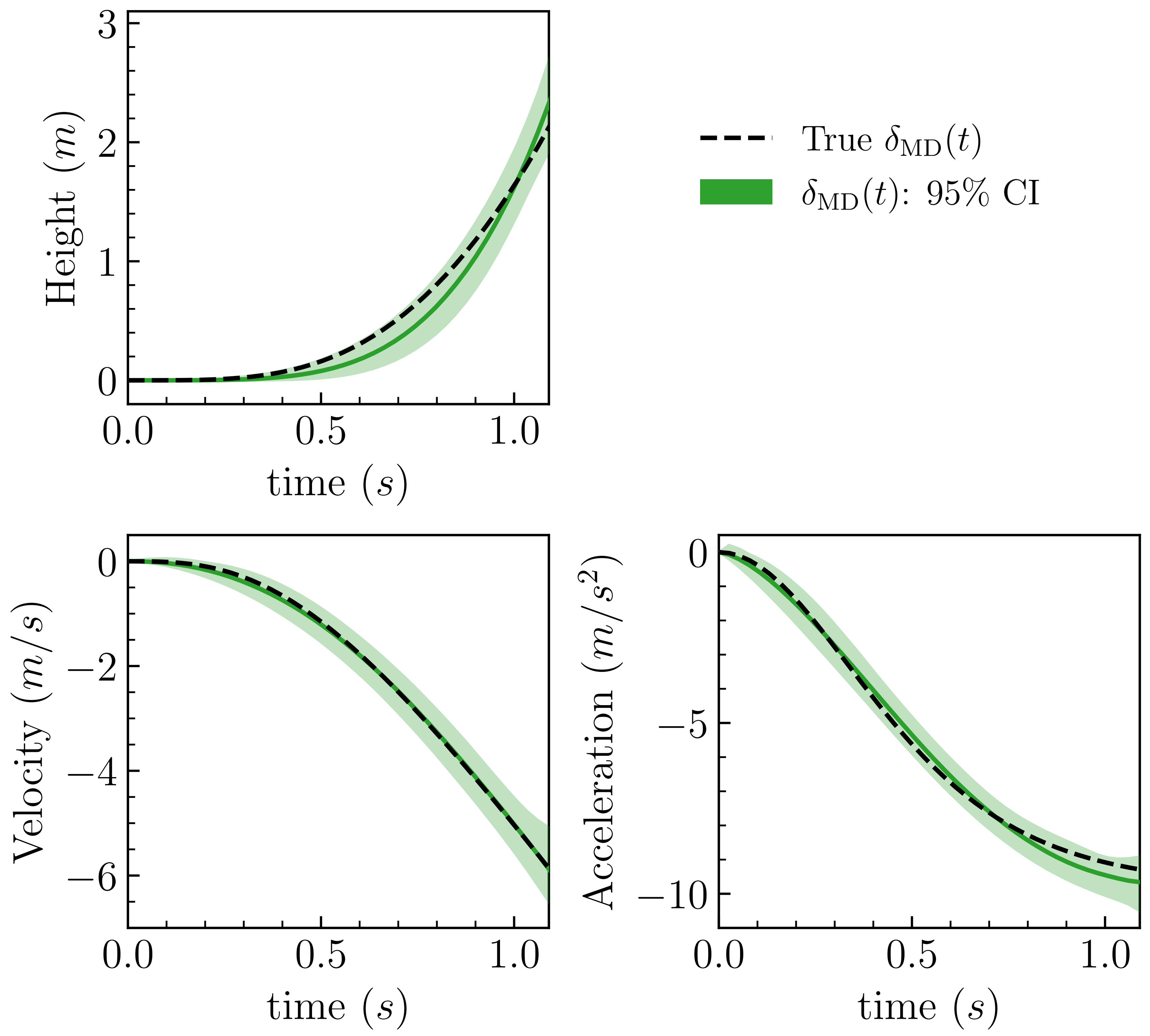
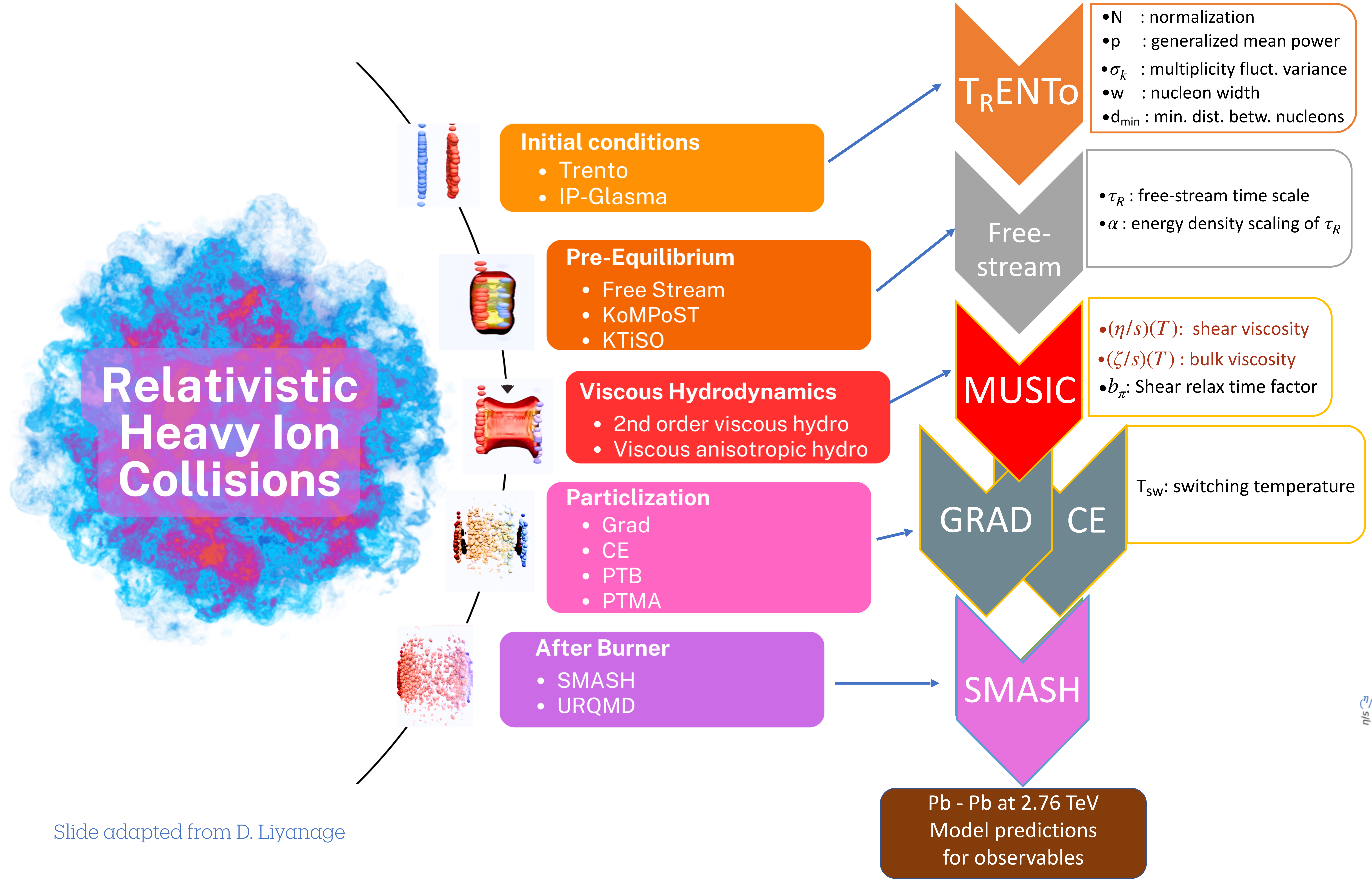


The learned discrepancy $\delta_{\rm MD}(x)$ informs where and how theory predictions fails.

Learning from data: Learn validity of theoretical models and infer physical properties of the system from data

Heavy-ion multi-stage physics model



JETSCAPE SIMS calibration

D. Everett *et al.* 2010.03928, 2011.01430

17 model parameters

Parameter name	Symbol
Normalization	N
Generalized mean	p
Multiplicity fluctuation	σ_k
Nucleon width	w
Min. dist. btw. nucleons	$d^3_{\rm min}$
Free-streaming time scale	τ_R
Free-streaming energy dep.	α
Temperature of (η/s) kink	$T_{\rm kink}$
Low temp. slope of (η/s)	$a_{\rm low}$
High temp. slope of (η/s)	$a_{\rm high}$
(η/s) at kink	$(\eta/s)_{\rm kink}$
Maximum of (ζ/s)	$(\zeta/s)_{\max}$
Temperature of (ζ/s) peak	T_ζ
Width of (ζ/s) peak	w_ζ
Asymmetry of (ζ/s) peak	λ_ζ
Shear relaxation time factor	b_π
Particlization temperature	$T_{\rm sw}$

$(\eta/s)(T)$

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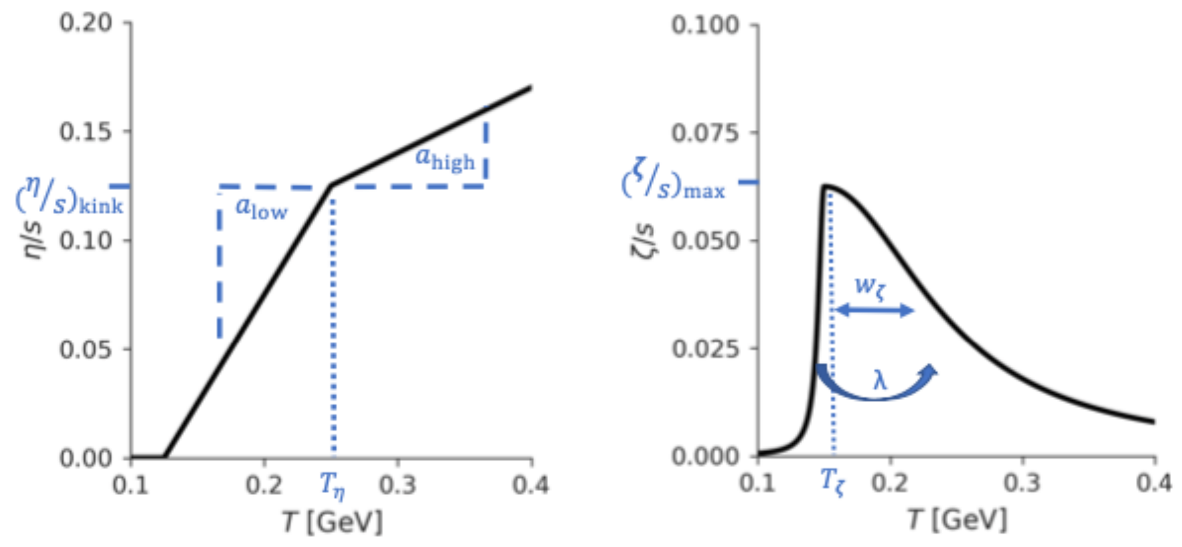

Slide adapted from D. Liyanage

Bayesian parameter inference: Pb+Pb collision at 2.76 TeV

[SJ, PLB 874, 140243 (2026)]

TRENTo → Free-streaming → 2nd order viscous hydro → Grad particlization → SMASH

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17 model parameters

$$\underbrace{y_{\exp}(x)}_{\text{Experimental observation}} = \underbrace{\eta_{\text{mod}}(x,\theta)}_{\text{Predictions from approximate theory}} + \underbrace{\delta_{\text{MD}}(x)}_{\text{Error of approximate theory predictions}} + \underbrace{\epsilon(x)}_{\text{Observation error}}$$

x: centrality

Bayesian updating of knowledge

$$\mathrm{pr}(A|B,I) = \frac{\mathrm{pr}(B|A,I)\mathrm{pr}(A|I)}{\mathrm{pr}(B|I)} \Longrightarrow \underbrace{\mathrm{pr}(\boldsymbol{\alpha}|\mathbf{y}_{\exp},I)}_{\text{posterior}} \propto \underbrace{\mathrm{pr}(\mathbf{y}_{\exp}|\boldsymbol{\alpha},I)}_{\text{likelihood}} \times \underbrace{\mathrm{pr}(\boldsymbol{\alpha}|I)}_{\text{prior}}$$

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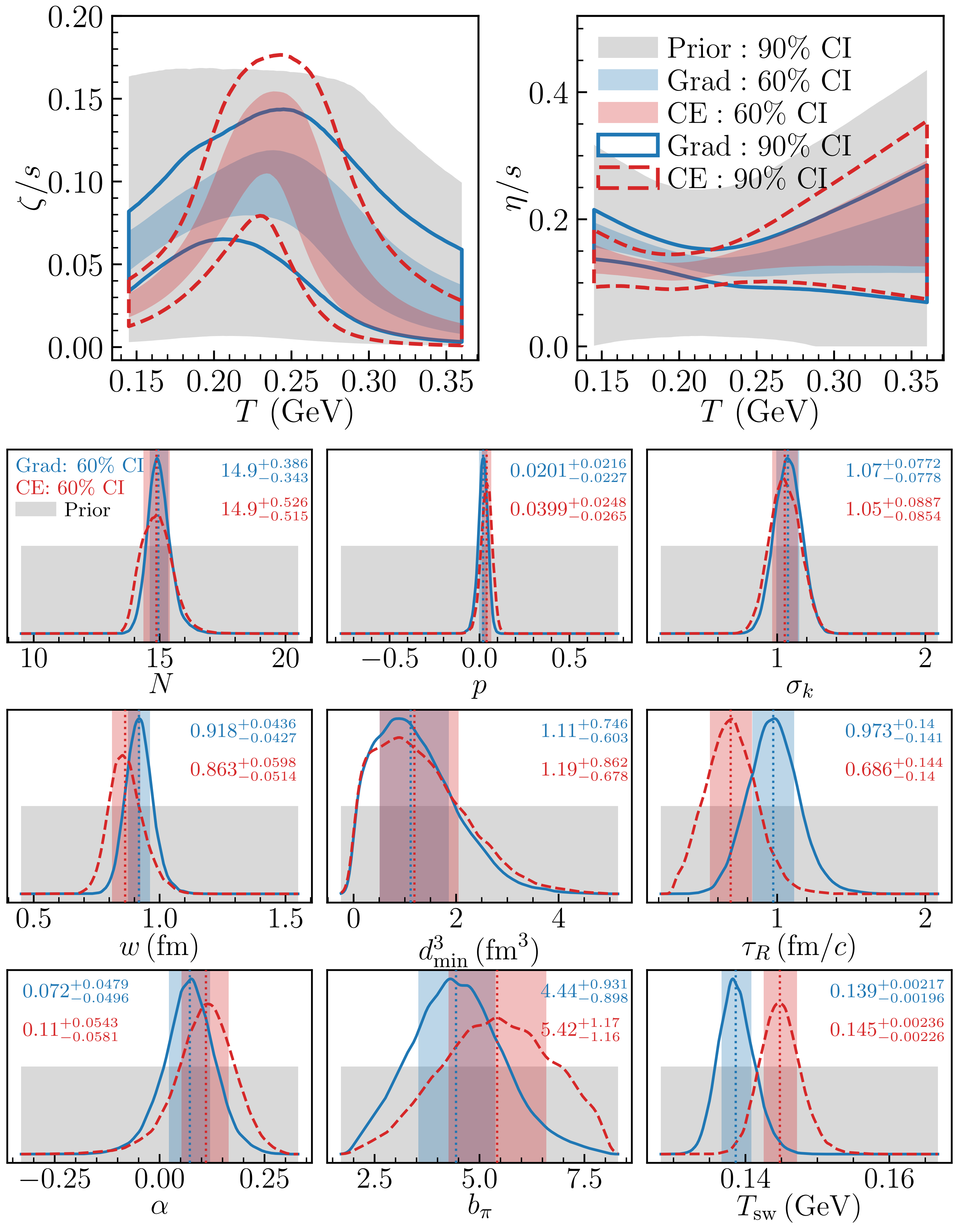

[D. Everett *et al.* 2010.03928, 2011.01430]

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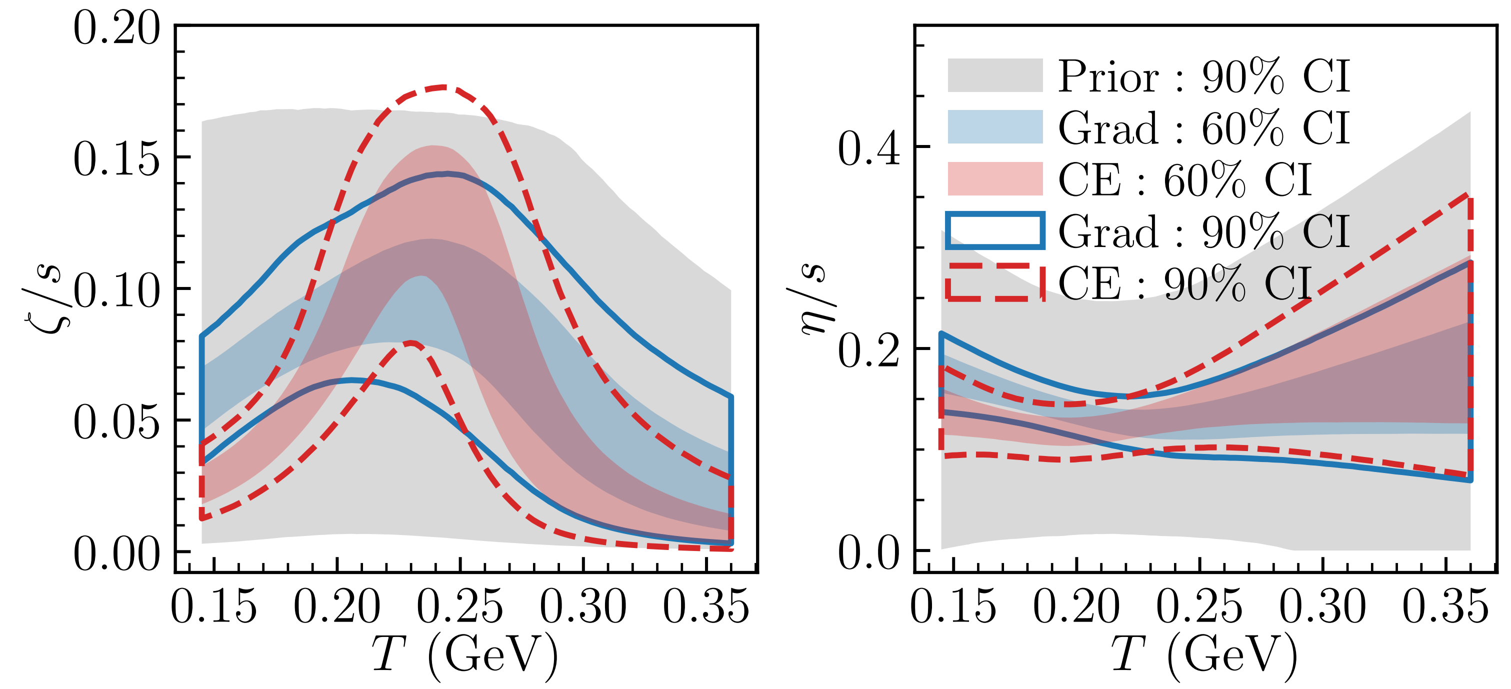

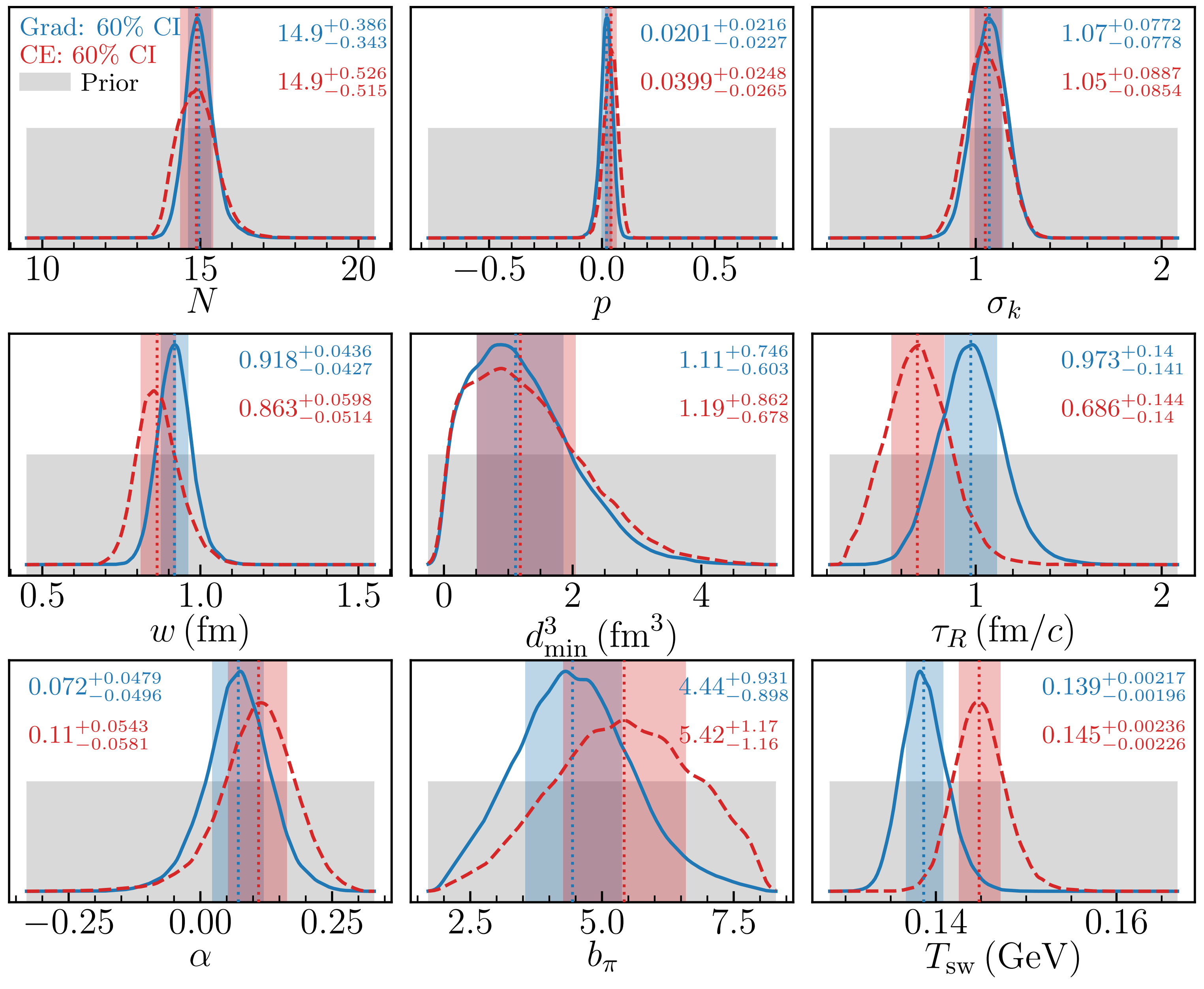

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Prior knowledge about theory domain (for both):
Multistage JETSCAPE model is more valid at small centralities (x) than at large centralities.
=> **Theory is more correct at small x than at large x**
=> $\delta_{\mathrm{MD}}(x)$ **should be non-decreasing with x**

$$K(x_i, x_j \mid \phi) \equiv s^2 + \bar{c}^2 (x_i x_j)^r \exp\left(-\frac{\|x_i - x_j\|^2}{2\ell^2}\right)$$

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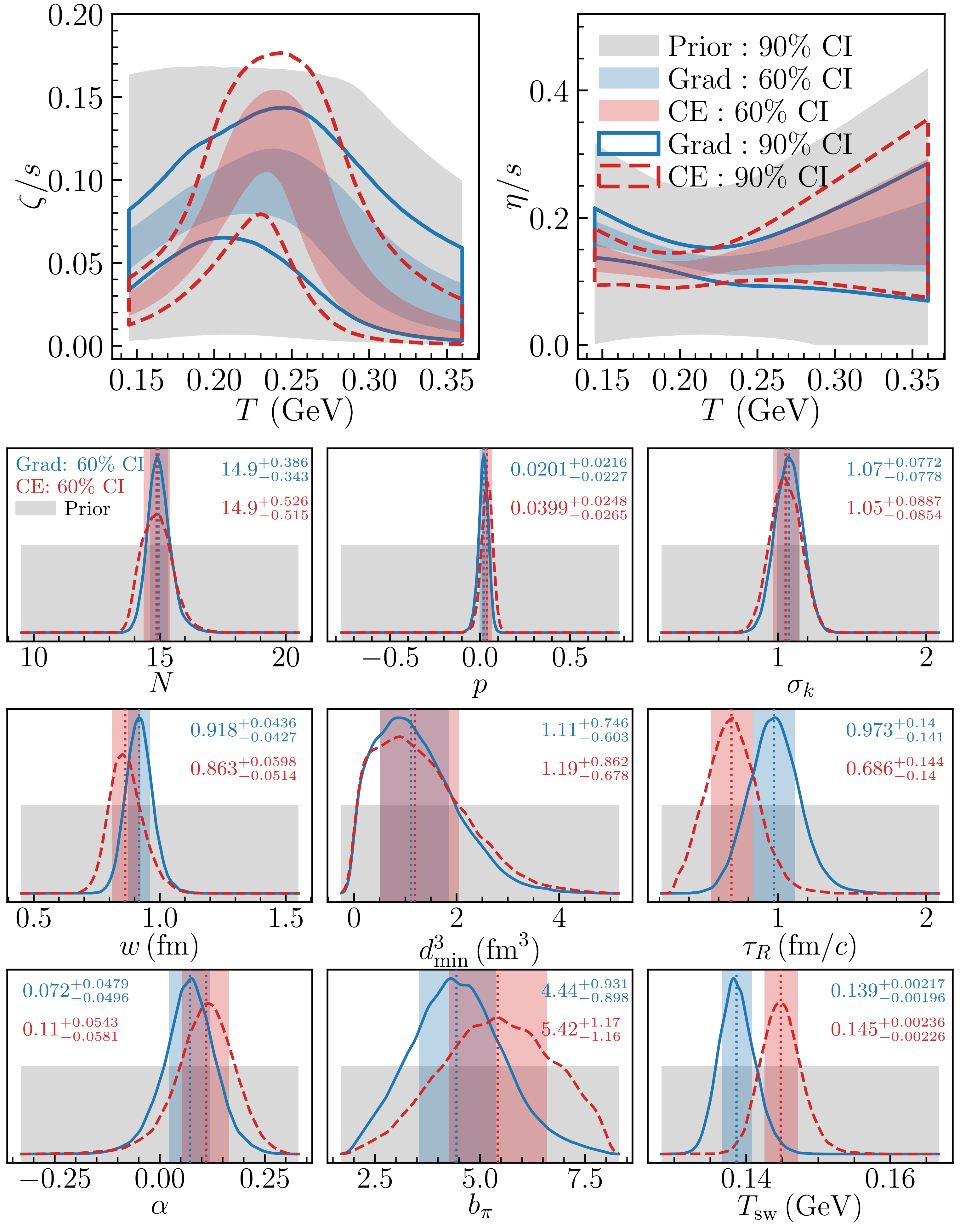

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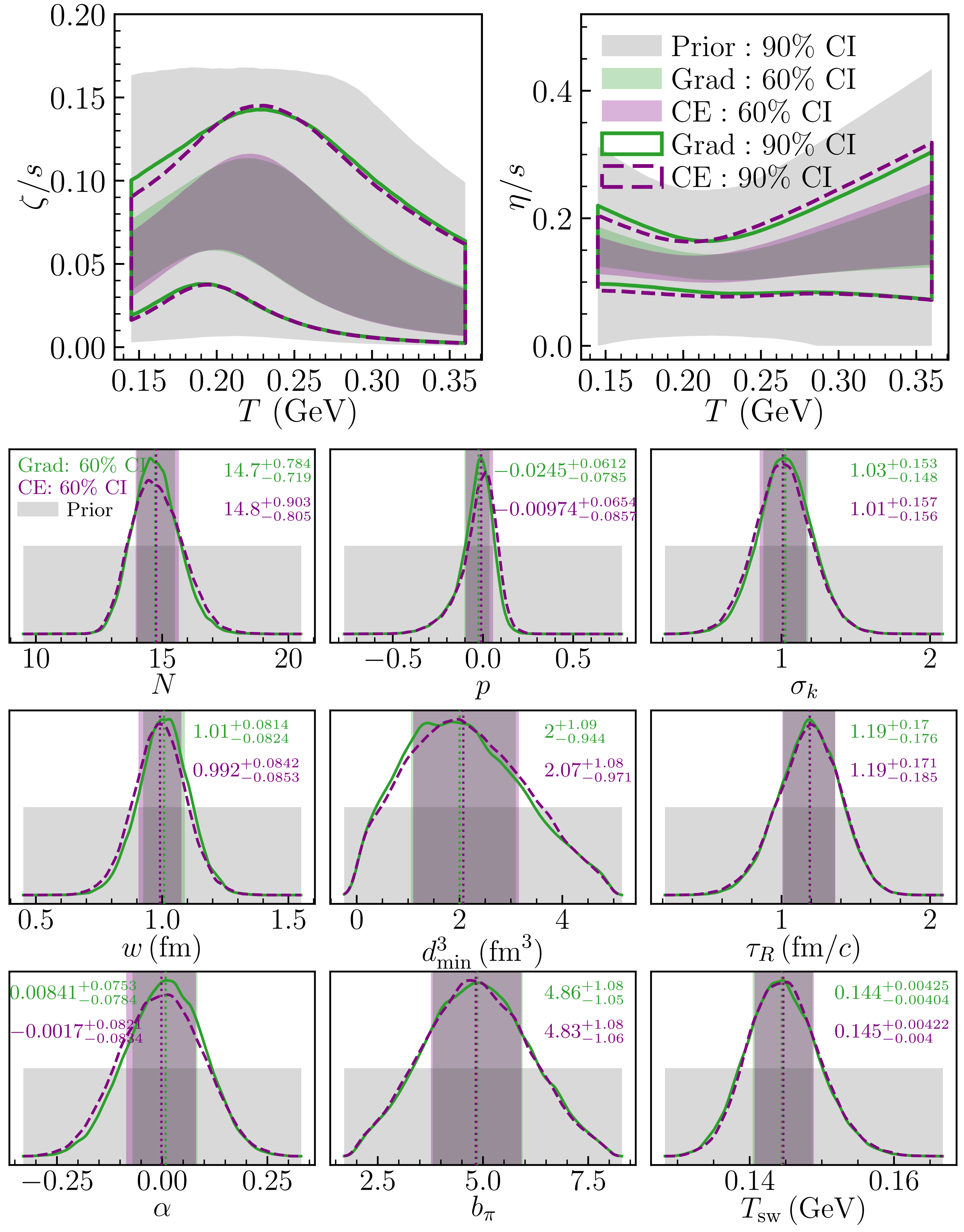
- Specific bulk viscosity posteriors differ between Grad (blue) and CE (red). Differences also visible for other parameters.
- Posteriors are almost indistinguishable for all parameters for Grad (green) and CE (purple).
- Error in theory predictions due to approximate or missing physics is quantified and does not propagate across stages.

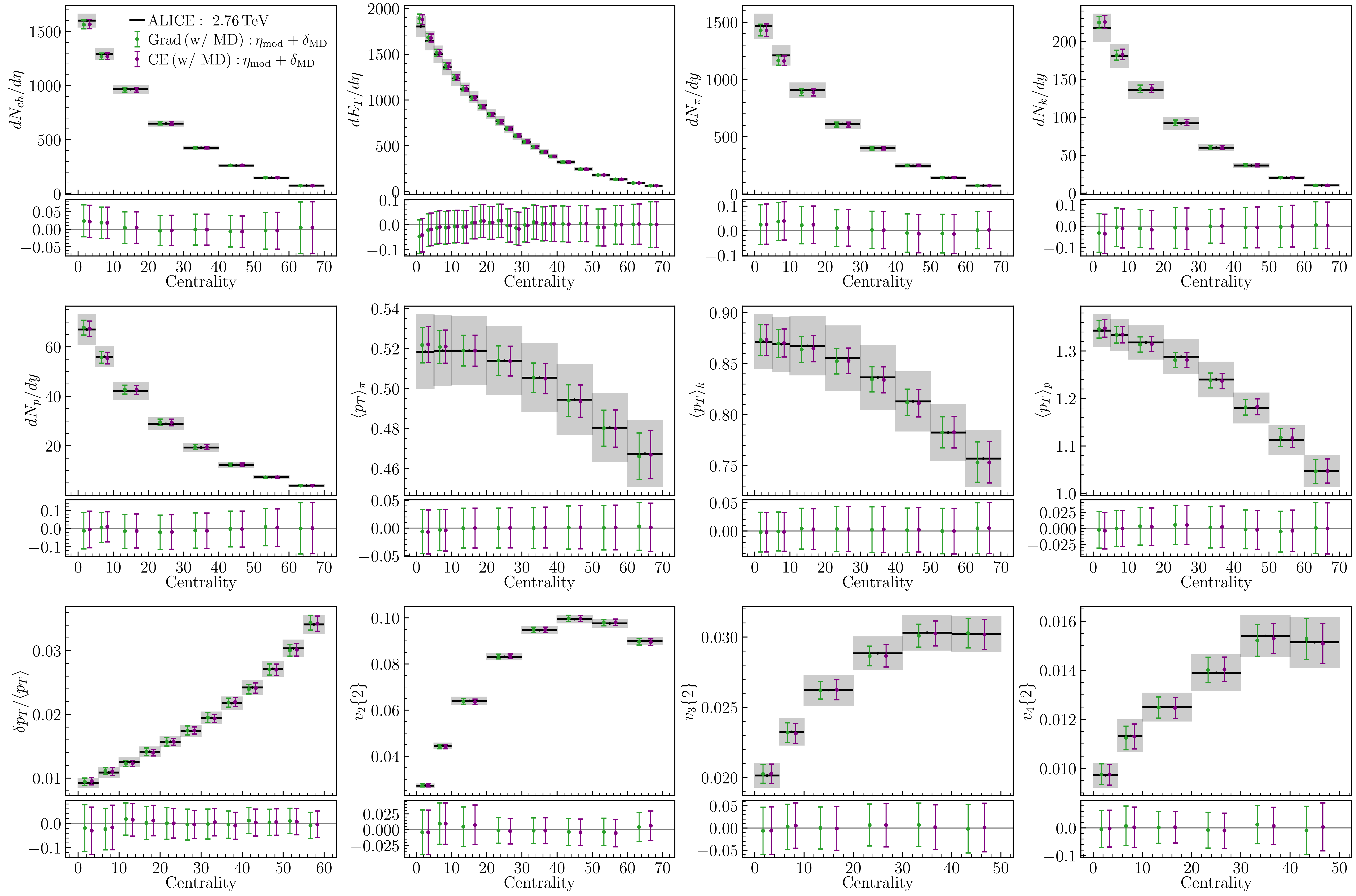
Accounting for theory uncertainties



[SJ, PLB 874, 140243 (2026)]

JETSCAPE Model + discrepancy predictions: Pb+Pb collision at 2.76 TeV

[SJ, PLB 874, 140243 (2026)]



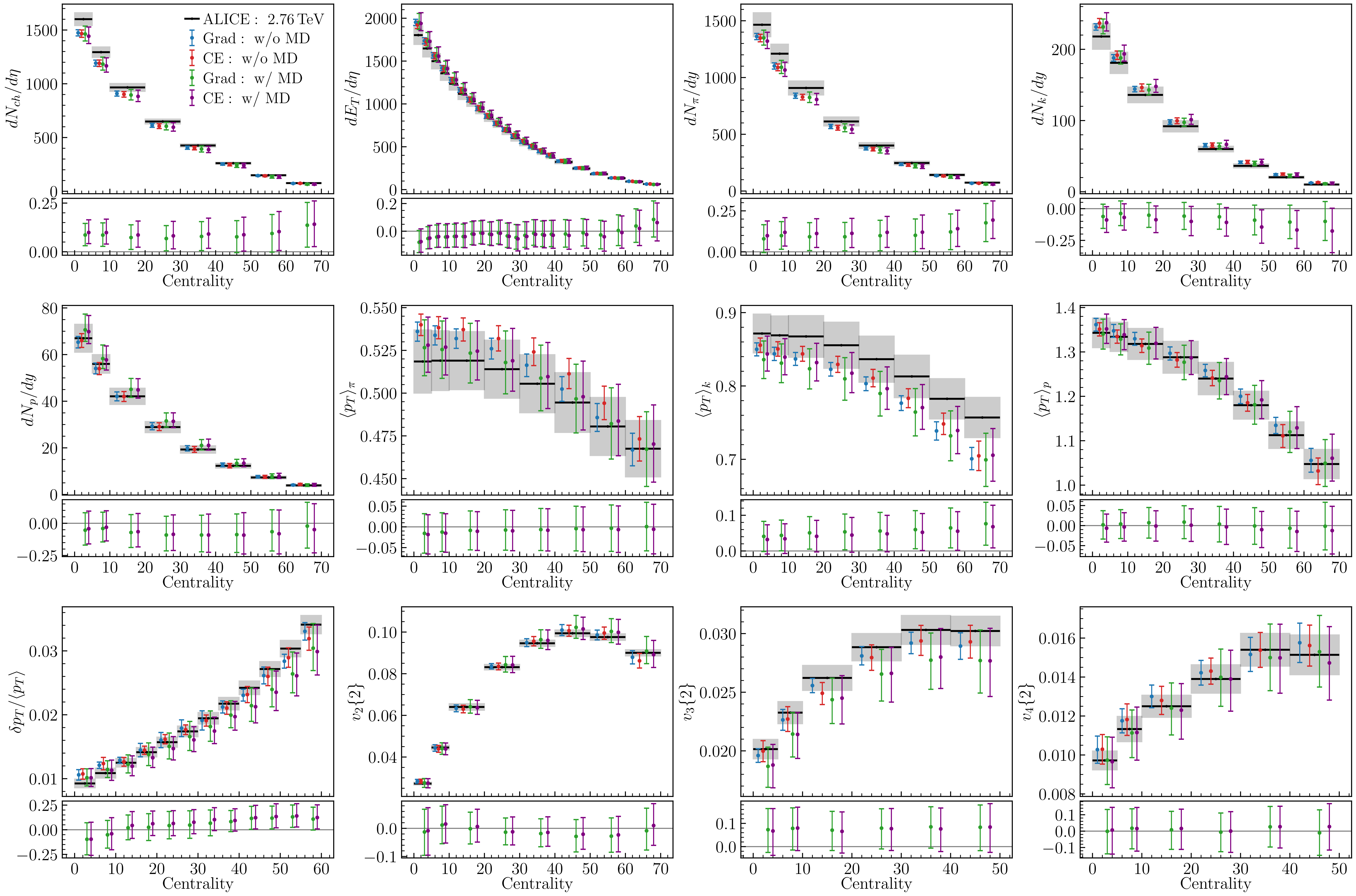
Predictions from
$\zeta(x;\theta,\phi) = \eta_{\mathrm{mod}}(x,\theta) + \delta_{\mathrm{MD}}(x,\phi)$

Excellent agreement with data.
Expected.

normalized error $(y_{\mathrm{exp}} - \eta_{\mathrm{mod}} - \delta_{\mathrm{MD}})/|\langle y_{\mathrm{exp}} \rangle|$

JETSCAPE model predictions: Pb+Pb collision at 2.76 TeV

[SJ, PLB 874, 140243 (2026)]



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- Both Grad and CE explains data equally good/bad.

normalized discrepancies $\underbrace{(y_{\text{exp}} - \eta_{\text{mod}})/|\langle y_{\text{exp}} \rangle|}_{\approx\, \delta_{\text{MD}}(x,\phi)}$

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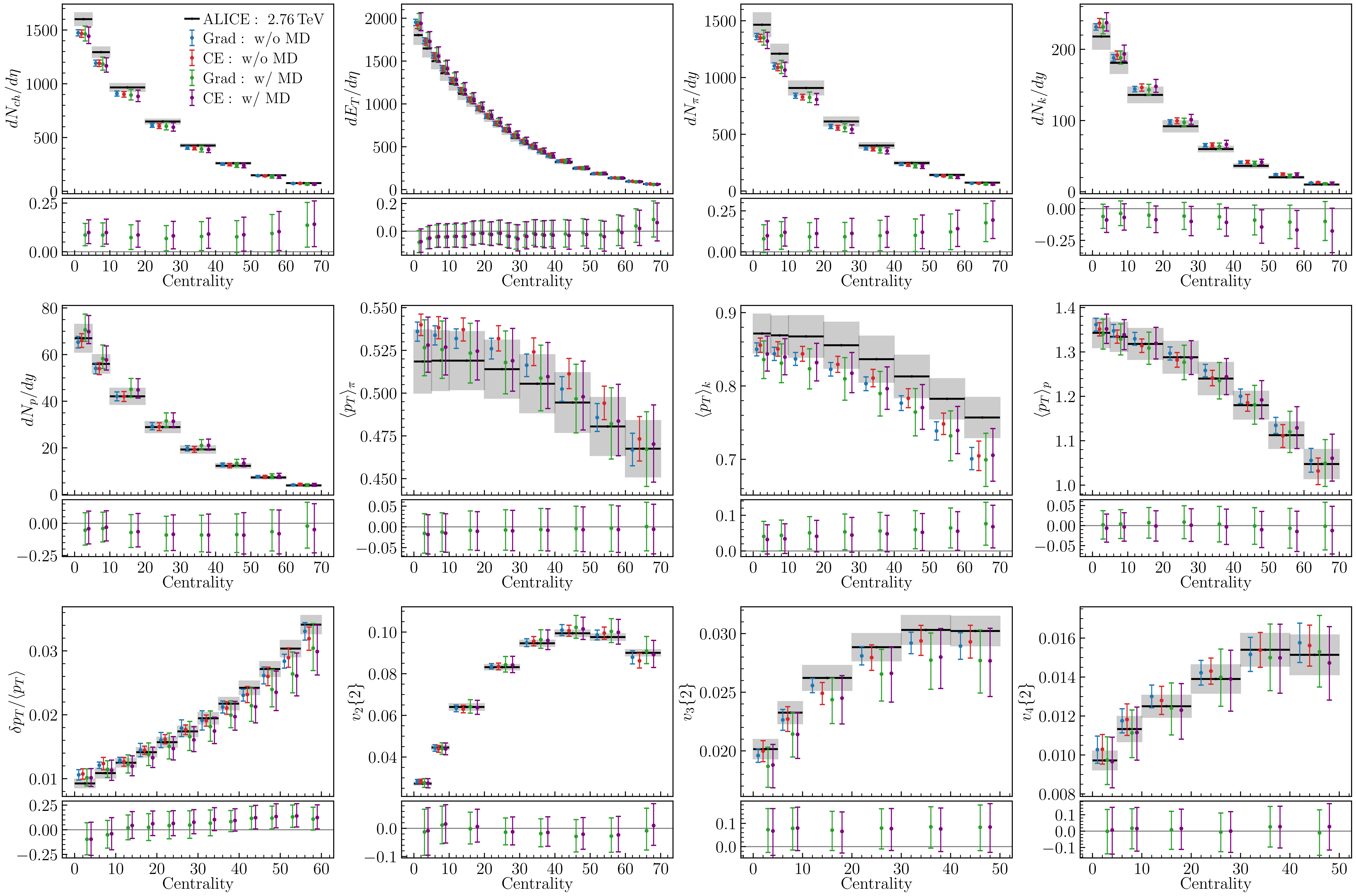

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- Both Grad and CE explains data equally good/bad.

- Theory predicts many observables correctly => learned $\delta_{\text{MD}}(x)$ are small for these.
- Deviations observed in w/ MD predictions => learned $\delta_{\text{MD}}(x)$.

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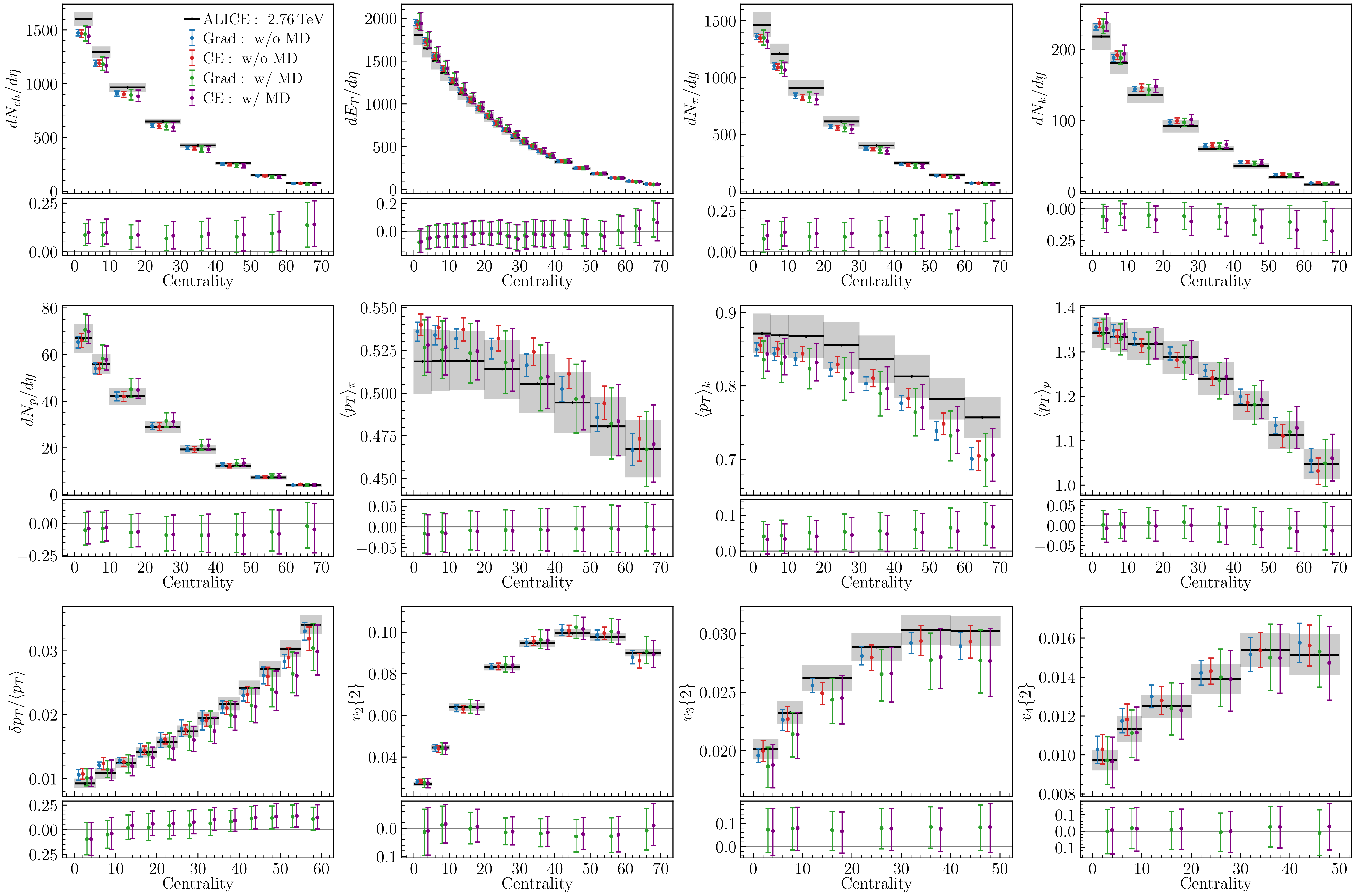

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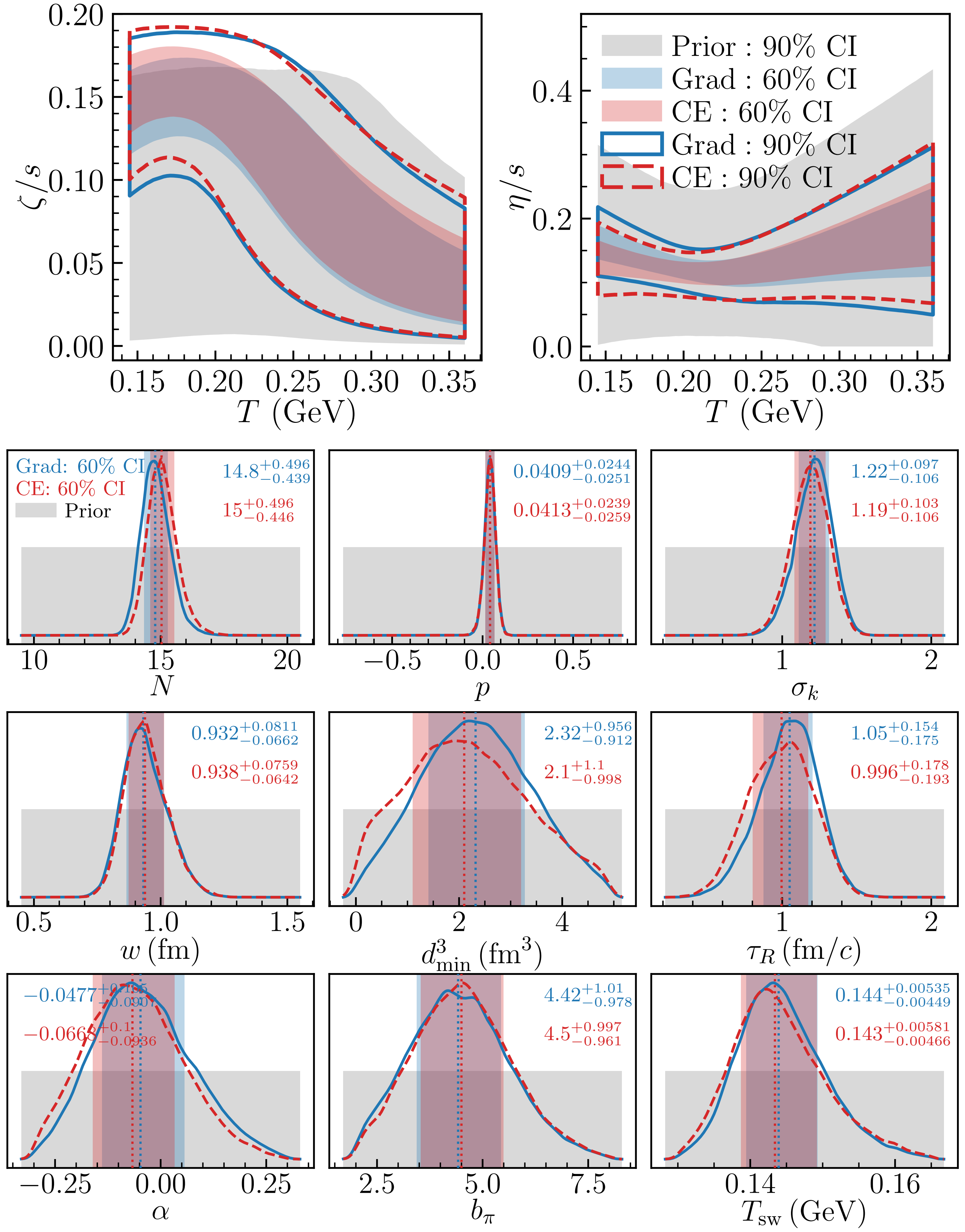
- Both Grad and CE explains data equally good/bad.
- Theory predicts many observables correctly => learned $\delta_{\mathrm{MD}}(x)$ are small for these.
- Deviations observed in w/ MD predictions => learned $\delta_{\mathrm{MD}}(x)$.
- **Large discrepancy in $\delta \mathbf{p_T}/\langle \mathbf{p_T} \rangle$ and $\mathbf{v_3\{2\}}$ indicates deficiencies in initial condition model.**

normalized discrepancies $\underbrace{(y_{\mathrm{exp}} - \eta_{\mathrm{mod}})/|\langle y_{\mathrm{exp}} \rangle|}_{\approx\, \delta_{\mathrm{MD}}(x,\phi)}$

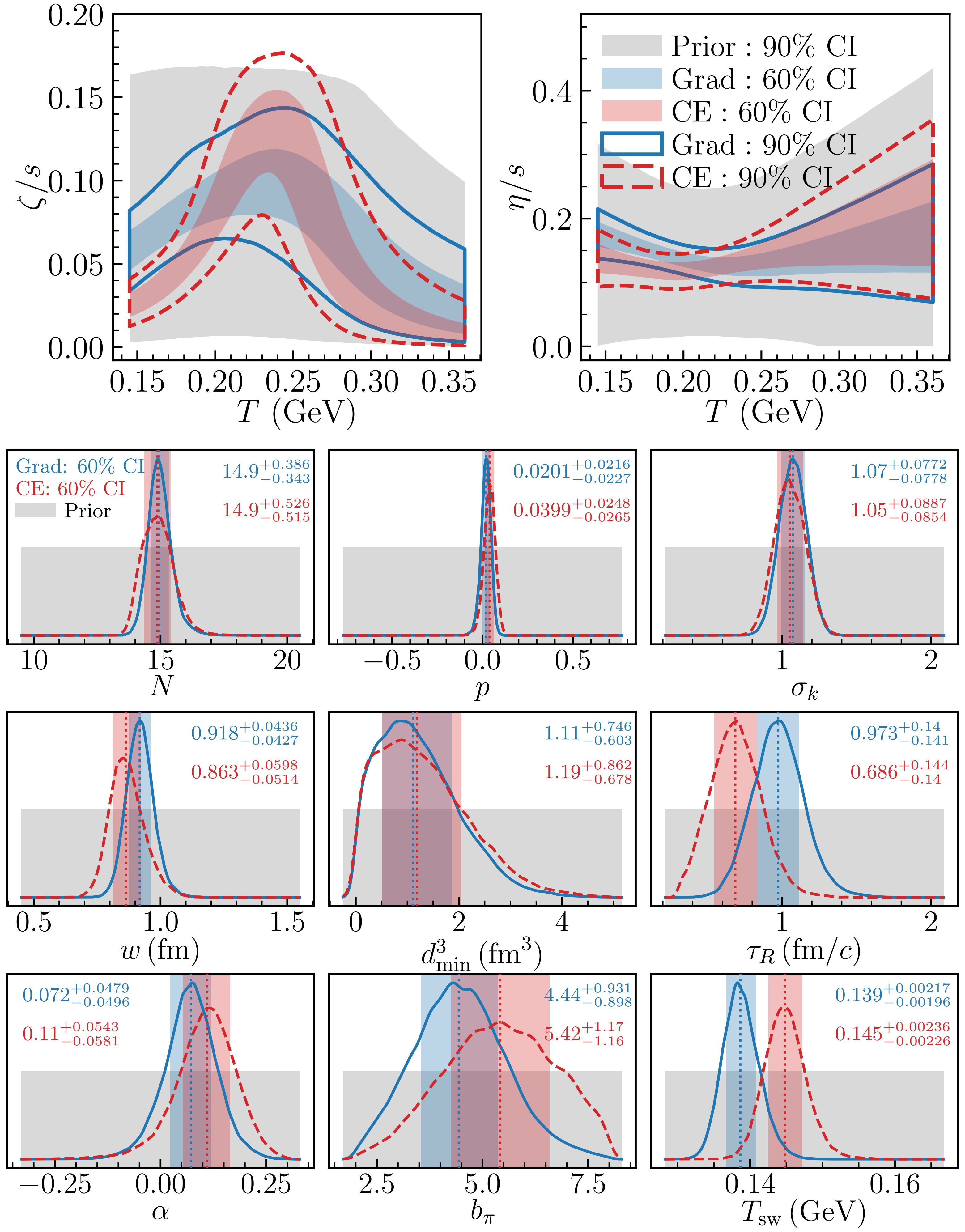
Bayesian inference on subset of observables : w/o MD

[SJ, PLB 874, 140243 (2026)]

Calibration with subset of observables: $dN_{\rm ch}/d\eta$, $dE_T/d\eta$, and $v_n\{2\}$ $(n = 2,3,4)$



Calibration with all observables



Posteriors for many parameters shift significantly when all measurements are included in the inference, indicating lack of robustness of the parameter estimate.

Bayesian inference on subset of observables : w/ MD

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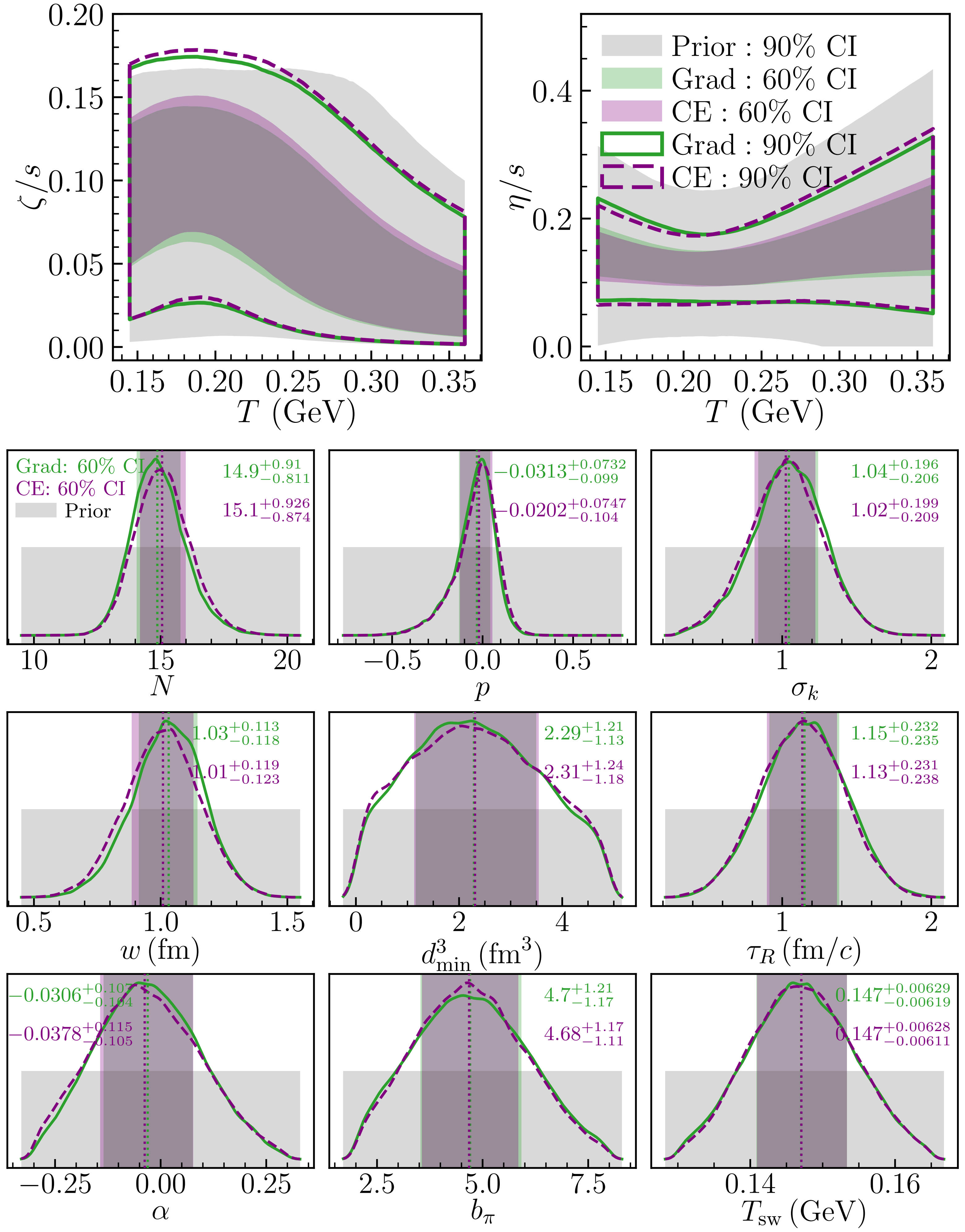
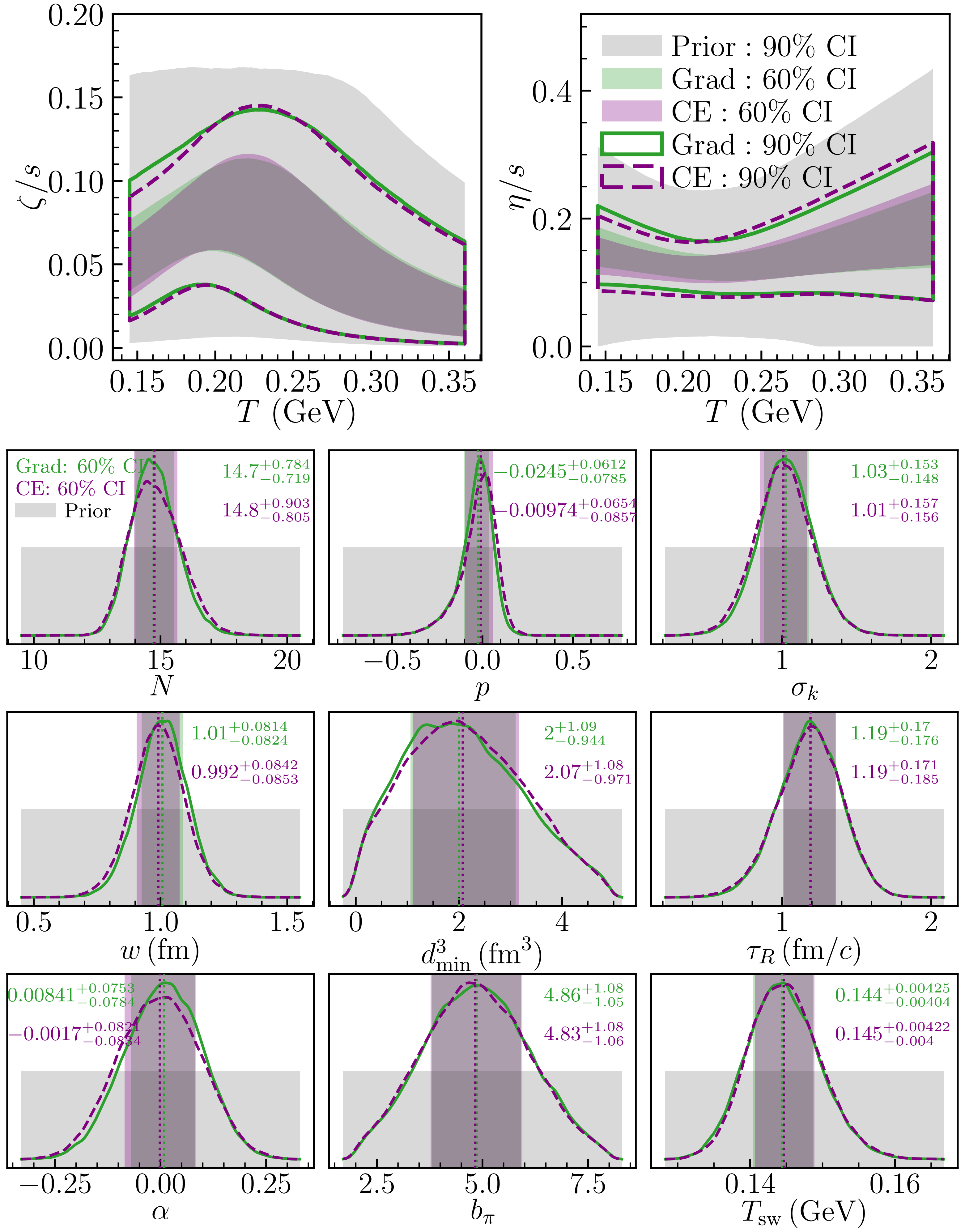


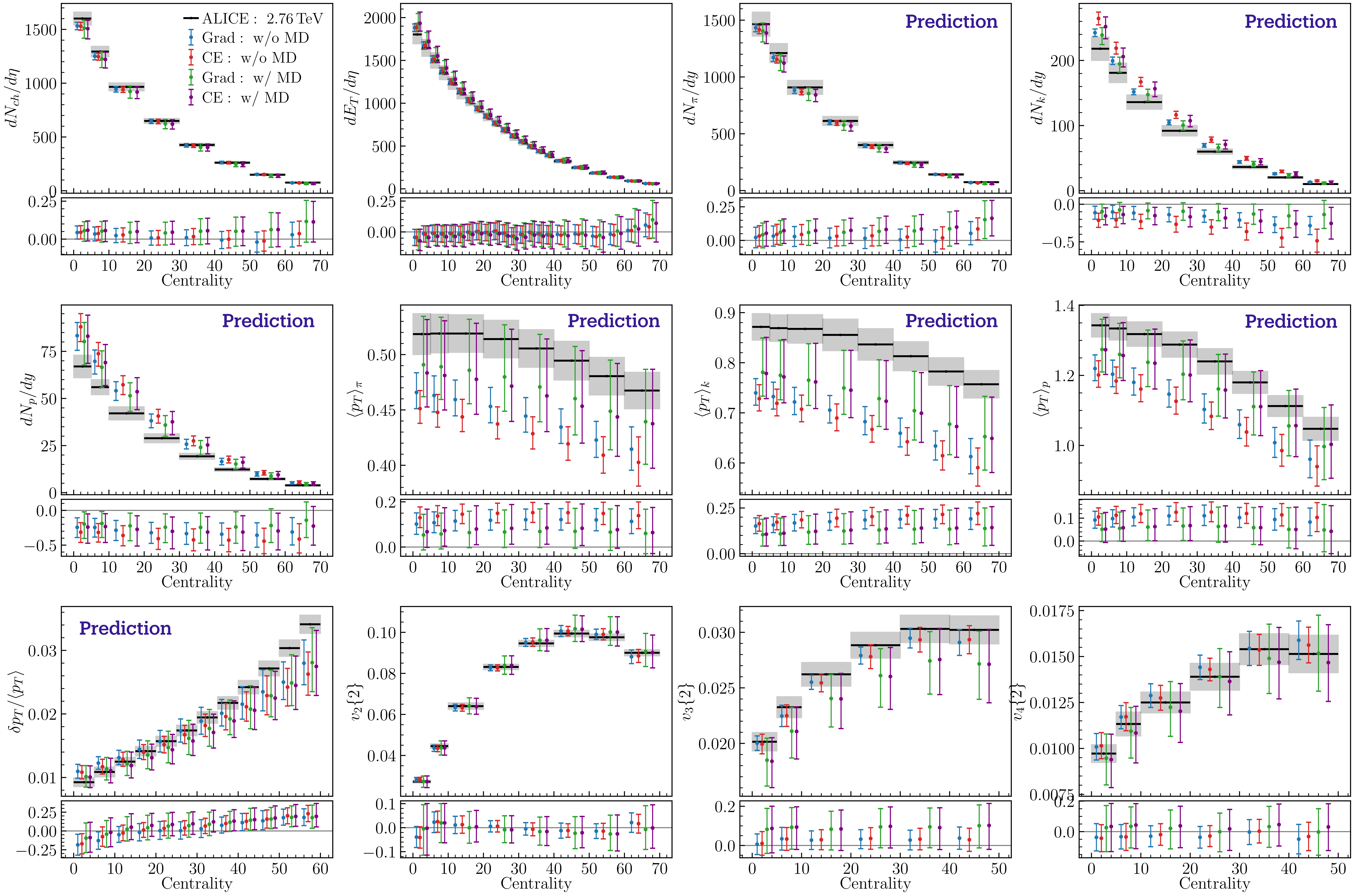
Calibration with all observables



Posteriors tighten without shifting, indicating robust parameter inference.

Model predictions: calibrated on subset of observables

[SJ, PLB 874, 140243 (2026)]



Predictions from $\eta_{\text{mod}}(x, \theta)$.

- Observables not used in calibration are labeled “Prediction”.
- w/o MD predictions for $\langle p_T \rangle_i,\ i \in \{\pi, k, p\}$ are outside $68\ \%$ CI prediction interval, reflecting overfitting.
- w/ MD results show comparatively better agreement with data, albeit with large error bars.

normalized discrepancies $\underbrace{(y_{\text{exp}} - \eta_{\text{mod}})}_{\approx\, \delta_{\text{MD}}(x,\phi)}/|\langle y_{\text{exp}} \rangle|$

Summary

- Often the goal of model-data comparison is to extract physically meaningful model parameters, **not to fit data**.
- Quantifying inadequacies of theoretical models is **crucial** for correct inference of physical model parameters.
  ➡ Implicit assumption of the model being universally valid is usually made in model-to-data comparison. This can lead to incorrect and misleading parameter estimates.
- Discussed a Bayesian framework that explicitly quantifies errors arising from inadequacies of theoretical model by statistically modeling theory errors, guided by qualitative knowledge of the theory's domain of reliability.
- Framework is general and can be applied to various problems in science.
  Open source code is provided: https://github.com/sjaiswal-tifr/ModelDiscrepancy

Thank you!

Backup slides

Gaussian Process: Distribution over functions

Formal definition: A Gaussian process is a collection of random variables, any finite number of which also have a joint Gaussian distribution.

Consider a multivariate Gaussian distribution: $p(\mathbf{y} \mid \Sigma) \propto \exp\left(-\frac{1}{2}\mathbf{y}^T\Sigma^{-1}\mathbf{y}\right)$

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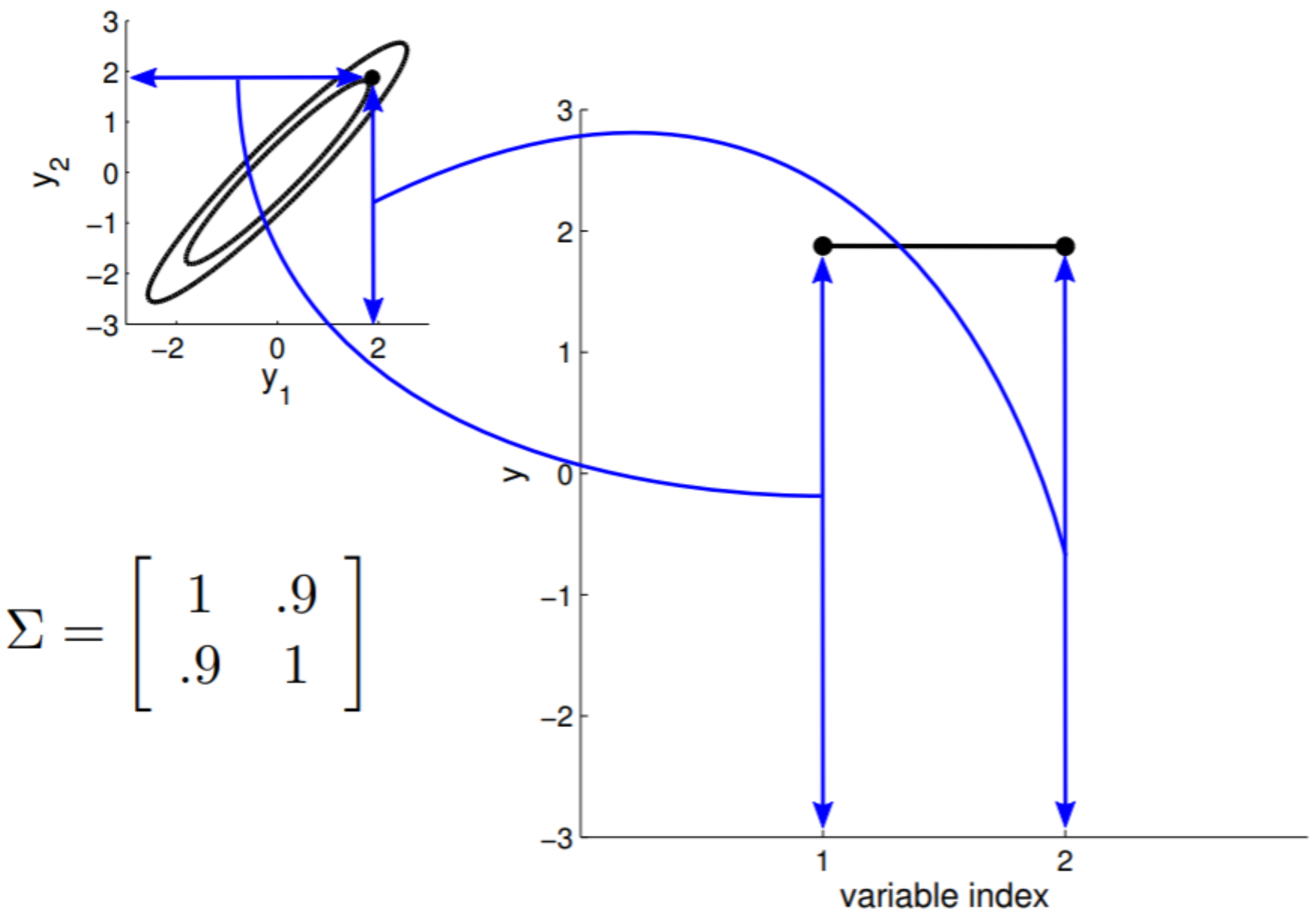
2D Gaussian distribution — A different representation



$$\Sigma = \begin{bmatrix} 1 & .9 \\ .9 & 1 \end{bmatrix}$$

https://thegradient.pub/gaussian-process-not-quite-for-dummies/

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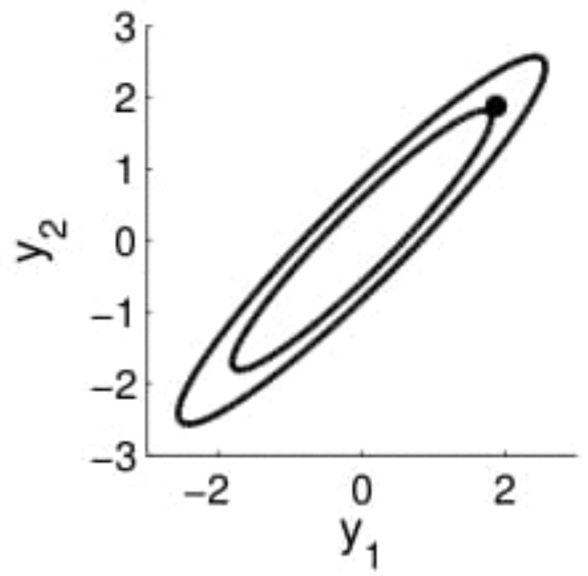

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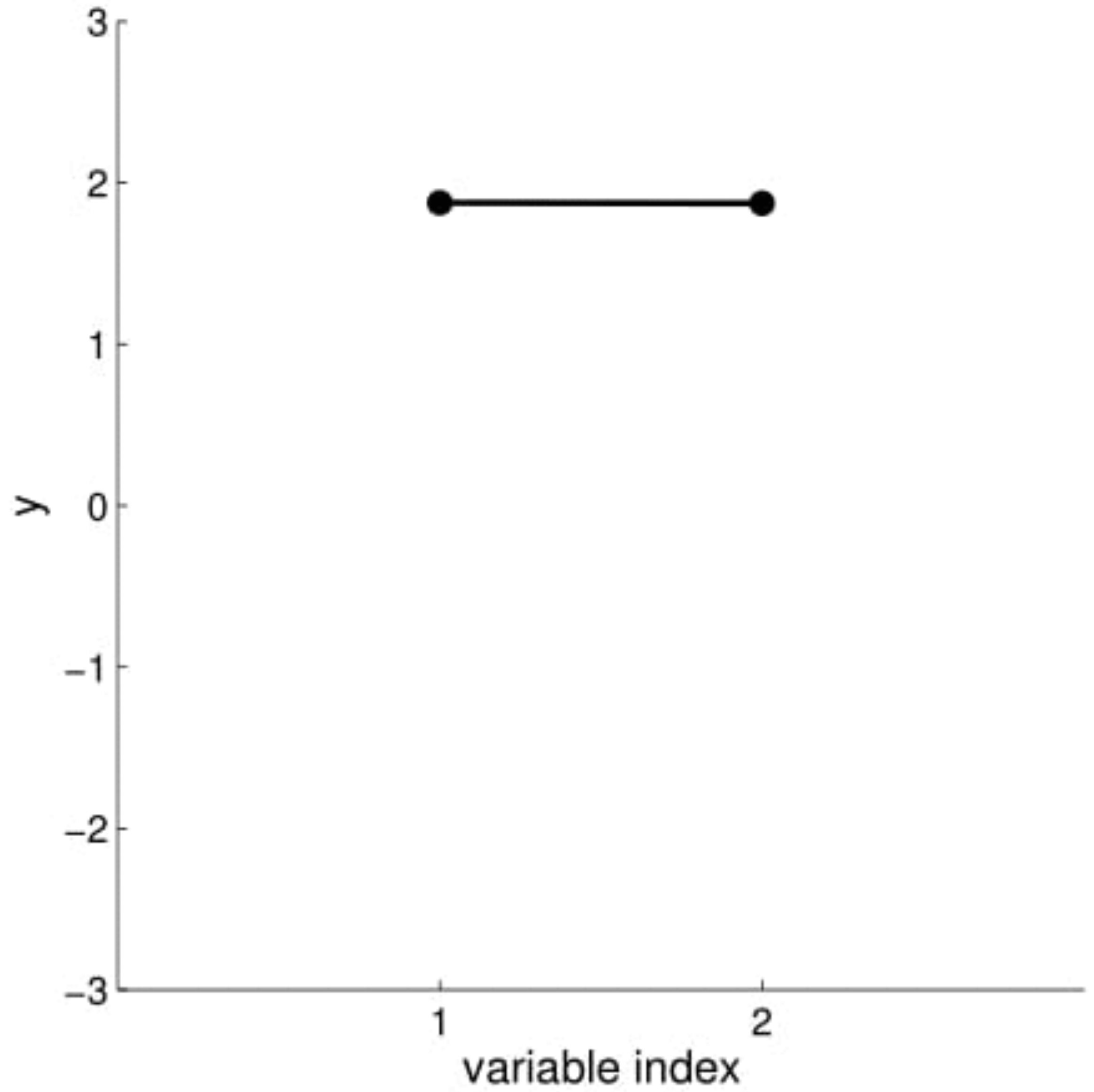

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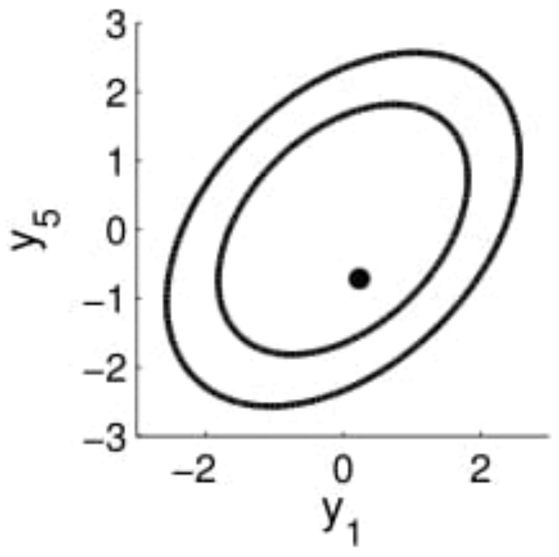

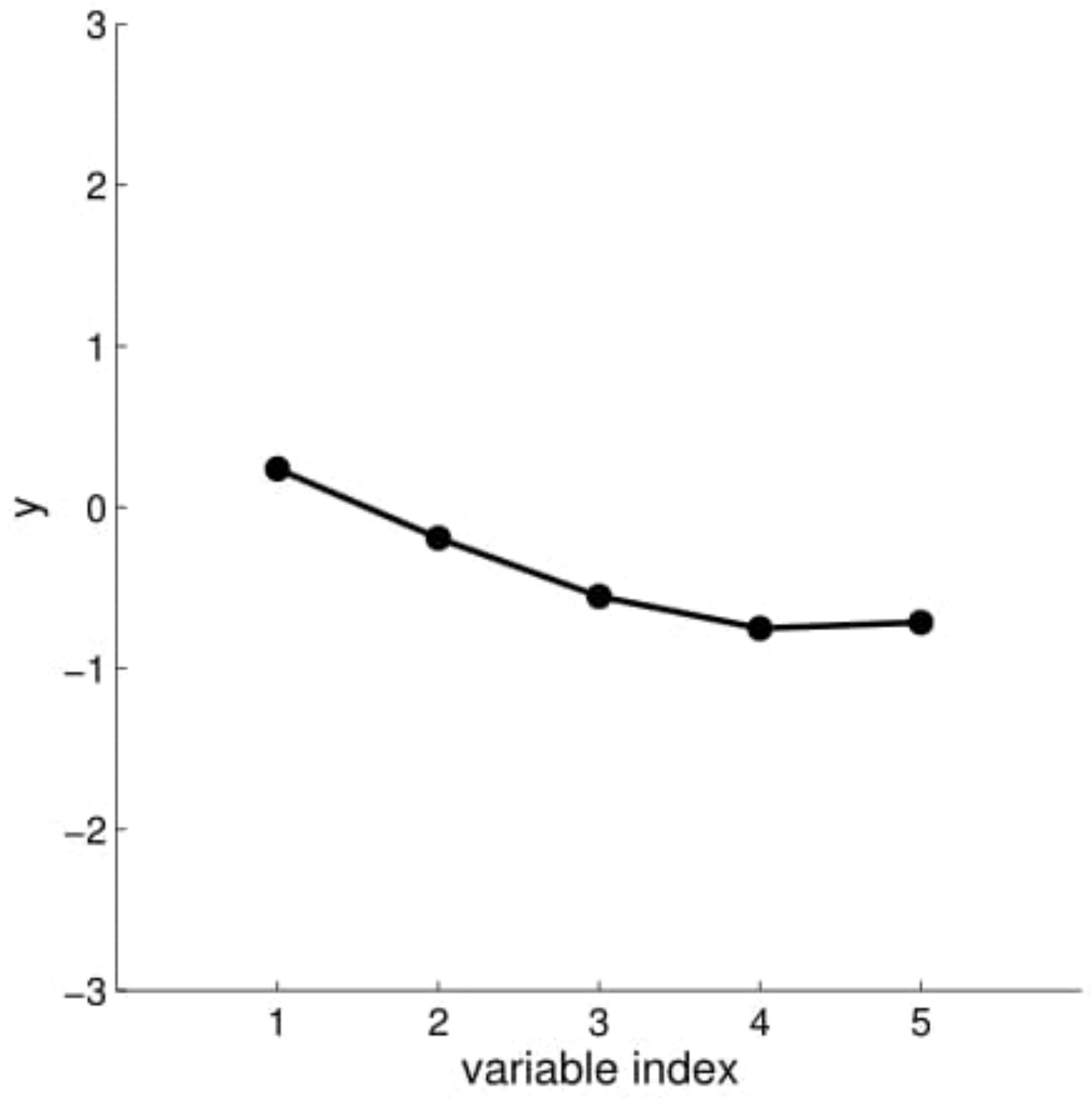

$$\Sigma = \begin{bmatrix} 1 & .9 & .8 & .6 & .4 \\ .9 & 1 & .9 & .8 & .6 \\ .8 & .9 & 1 & .9 & .8 \\ .6 & .8 & .9 & 1 & .9 \\ .4 & .6 & .8 & .9 & 1 \end{bmatrix}$$

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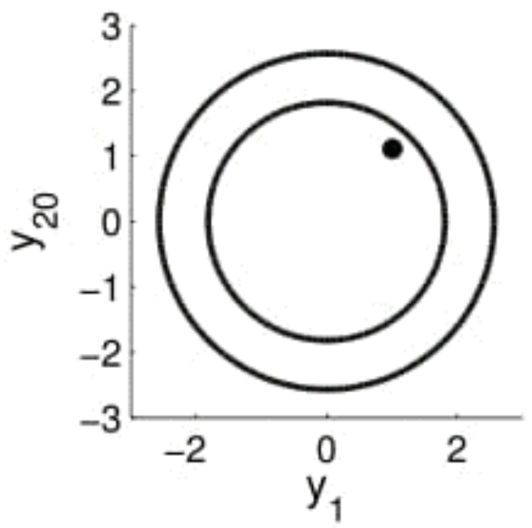

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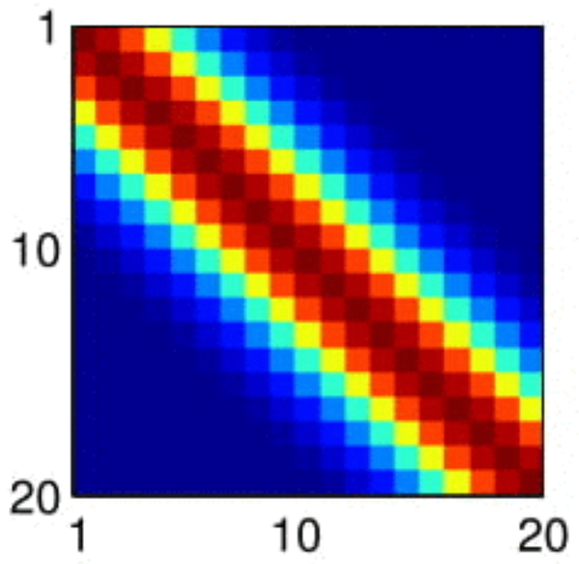

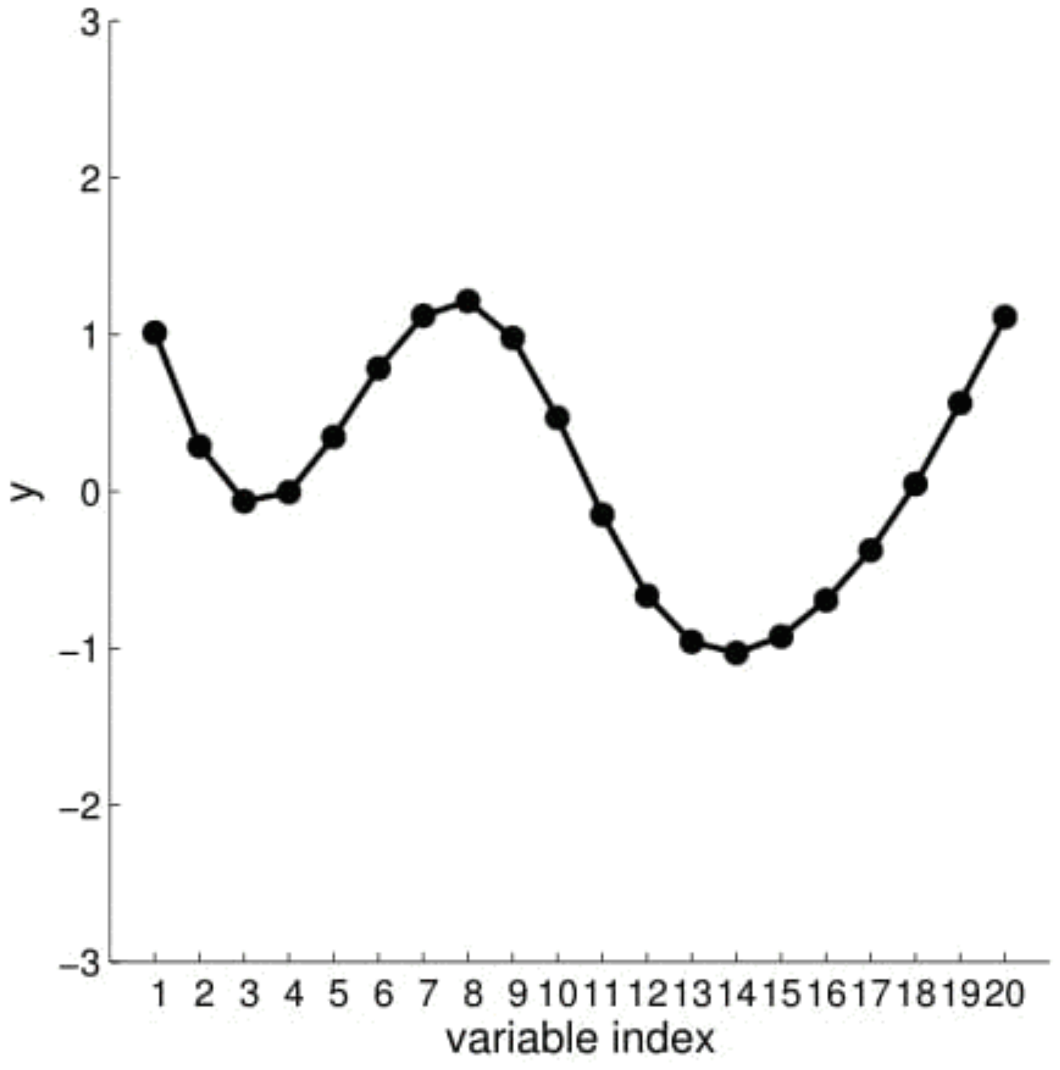

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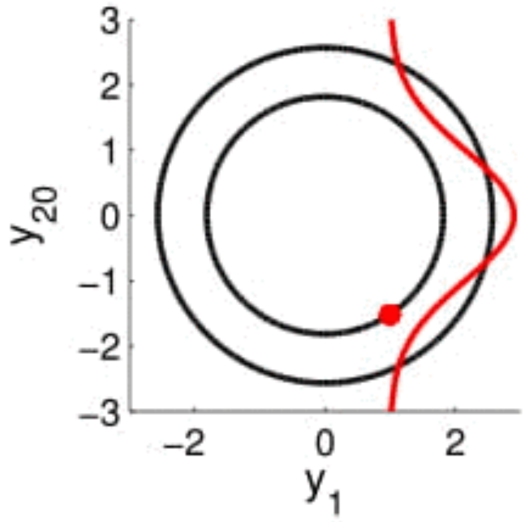
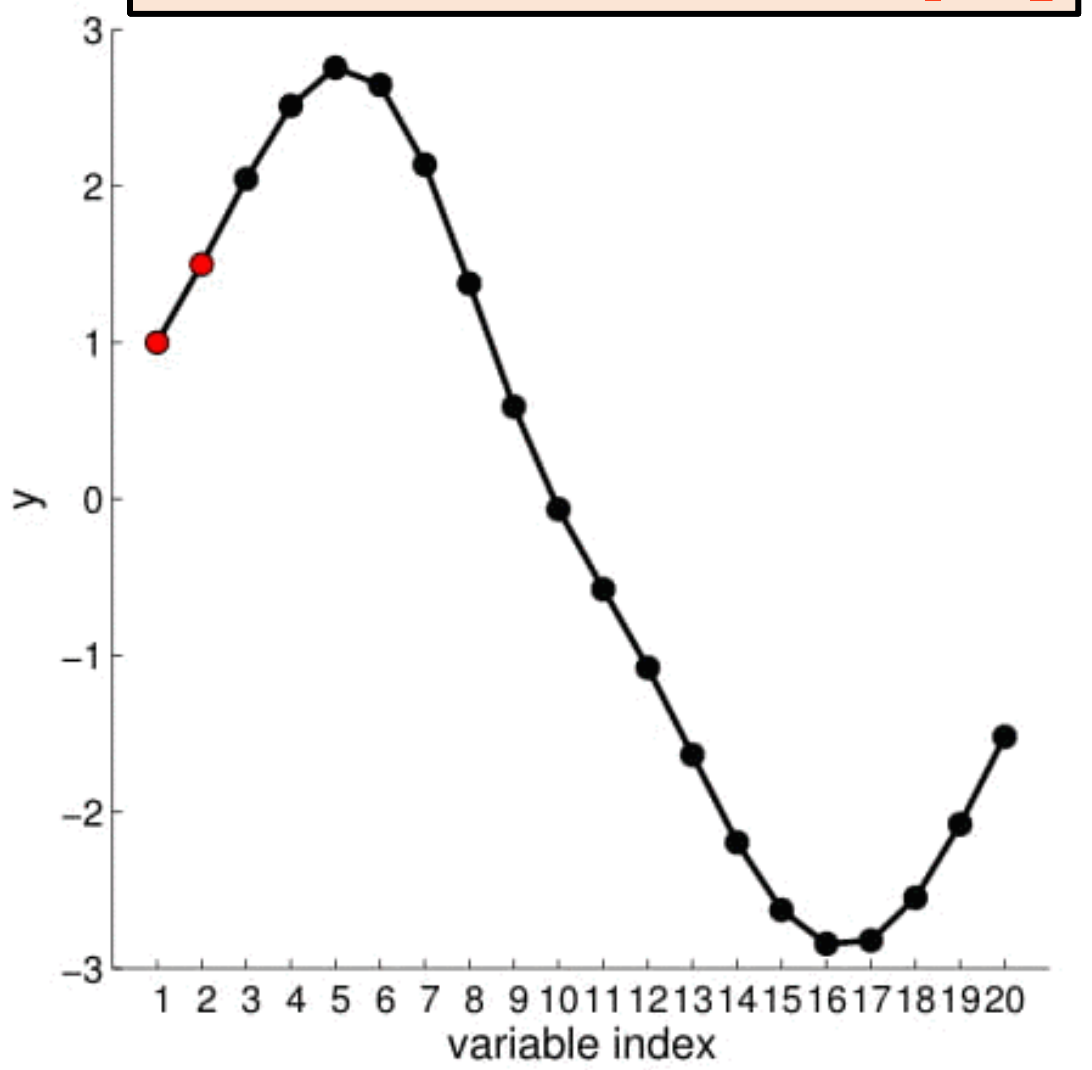


Conditioned on (fixing) $y_1,\ y_2$



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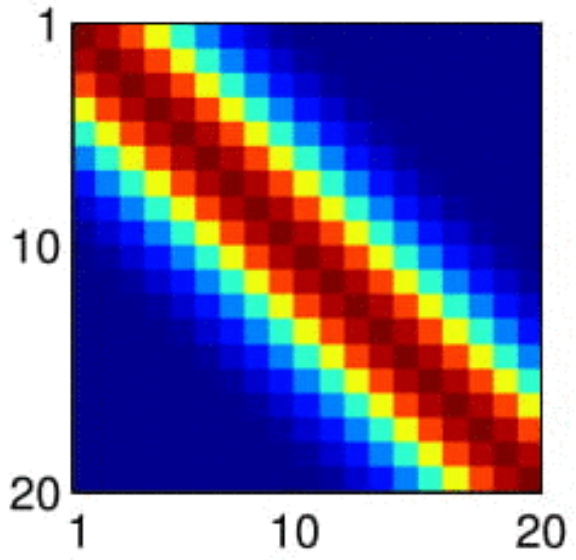

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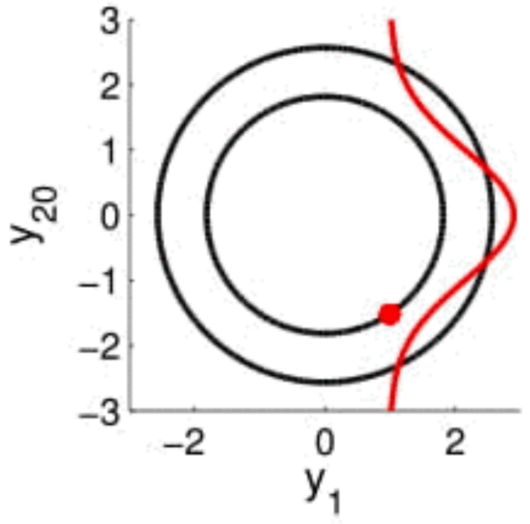

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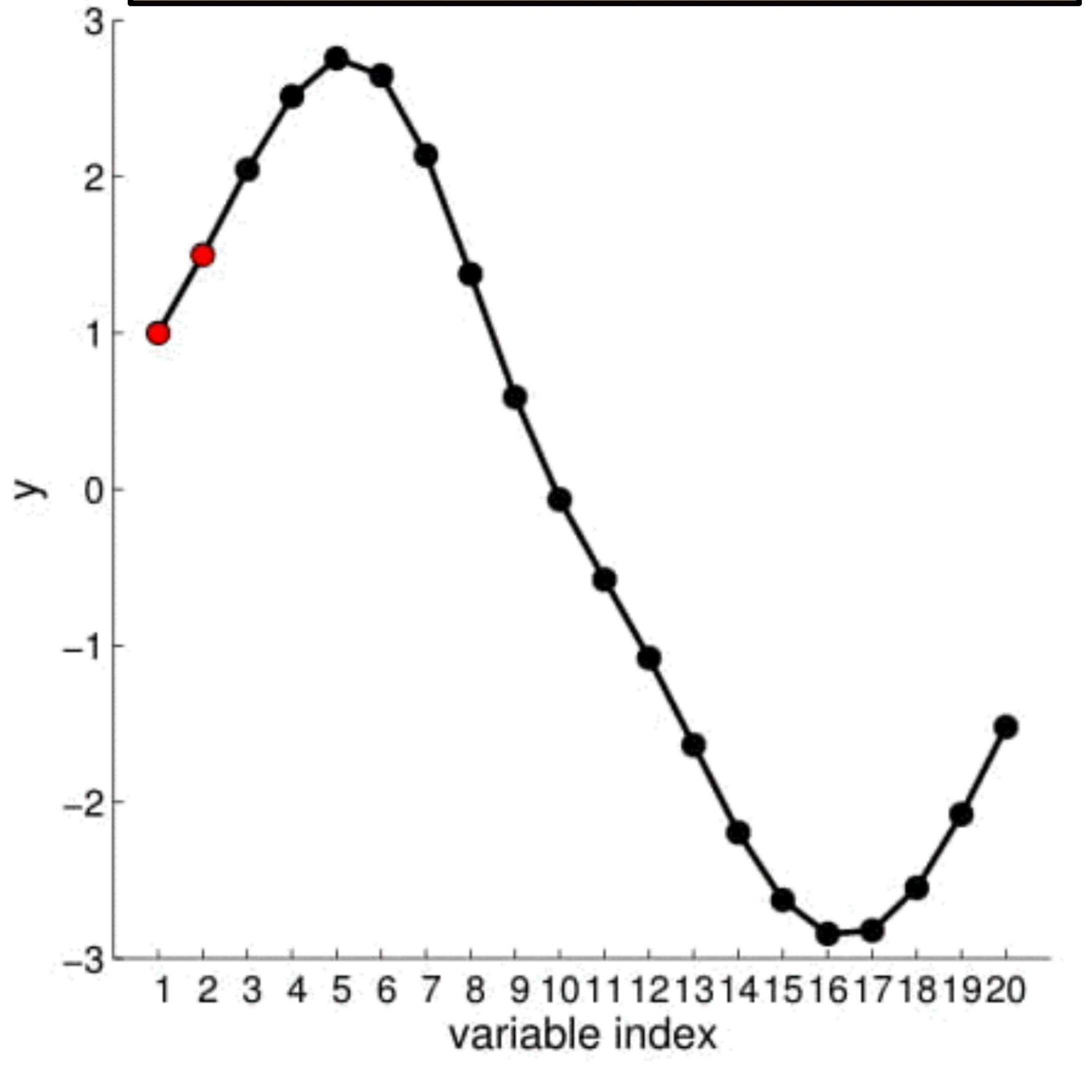

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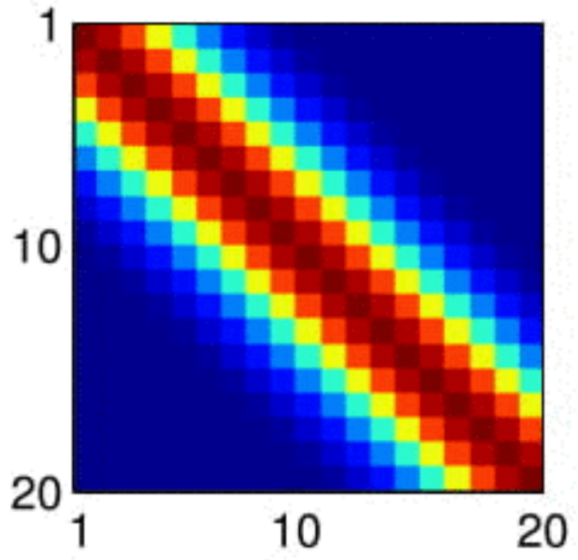
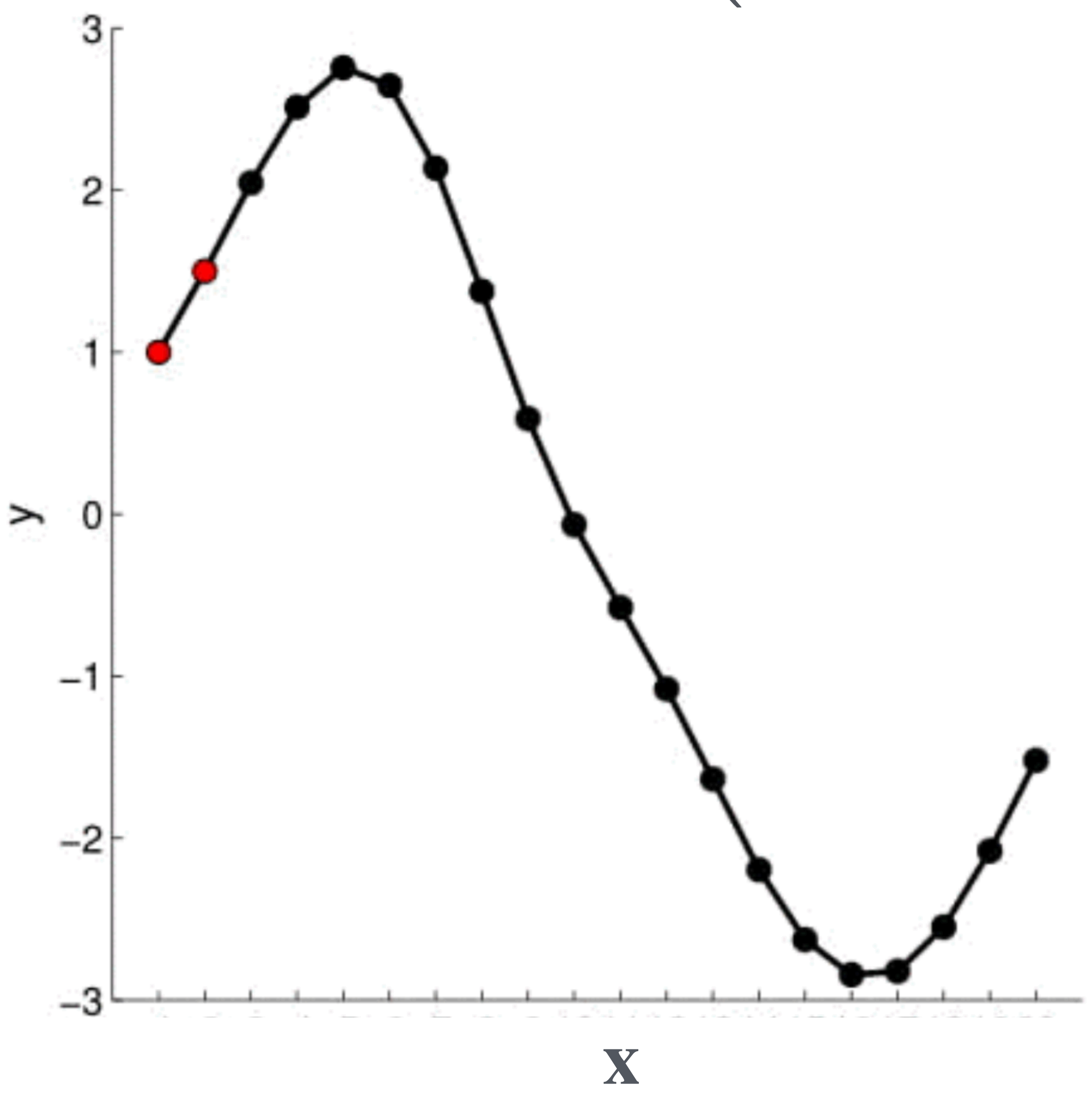


Computing Σ from the **covariance kernel**

$$\Sigma \equiv K(x_i, x_j) = \bar{c}^2 \exp\left(-\frac{\|x_i - x_j\|^2}{2\ell^2}\right)$$

$\mathbf{x}$ is a vector of length n ($= 20$ here)



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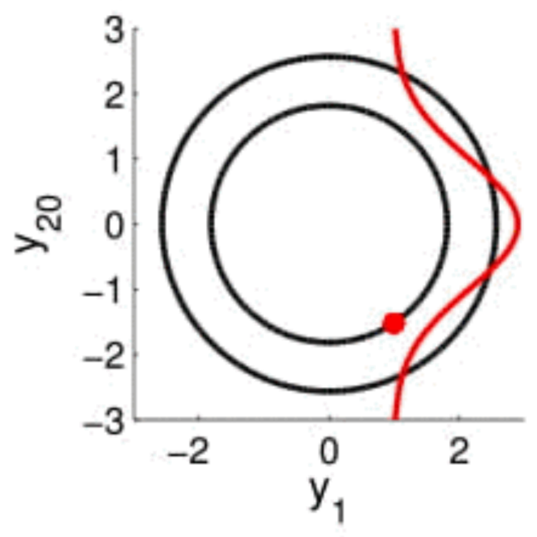

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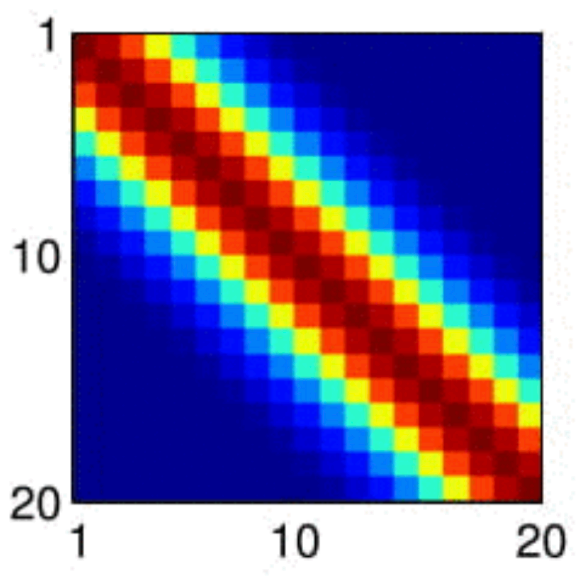

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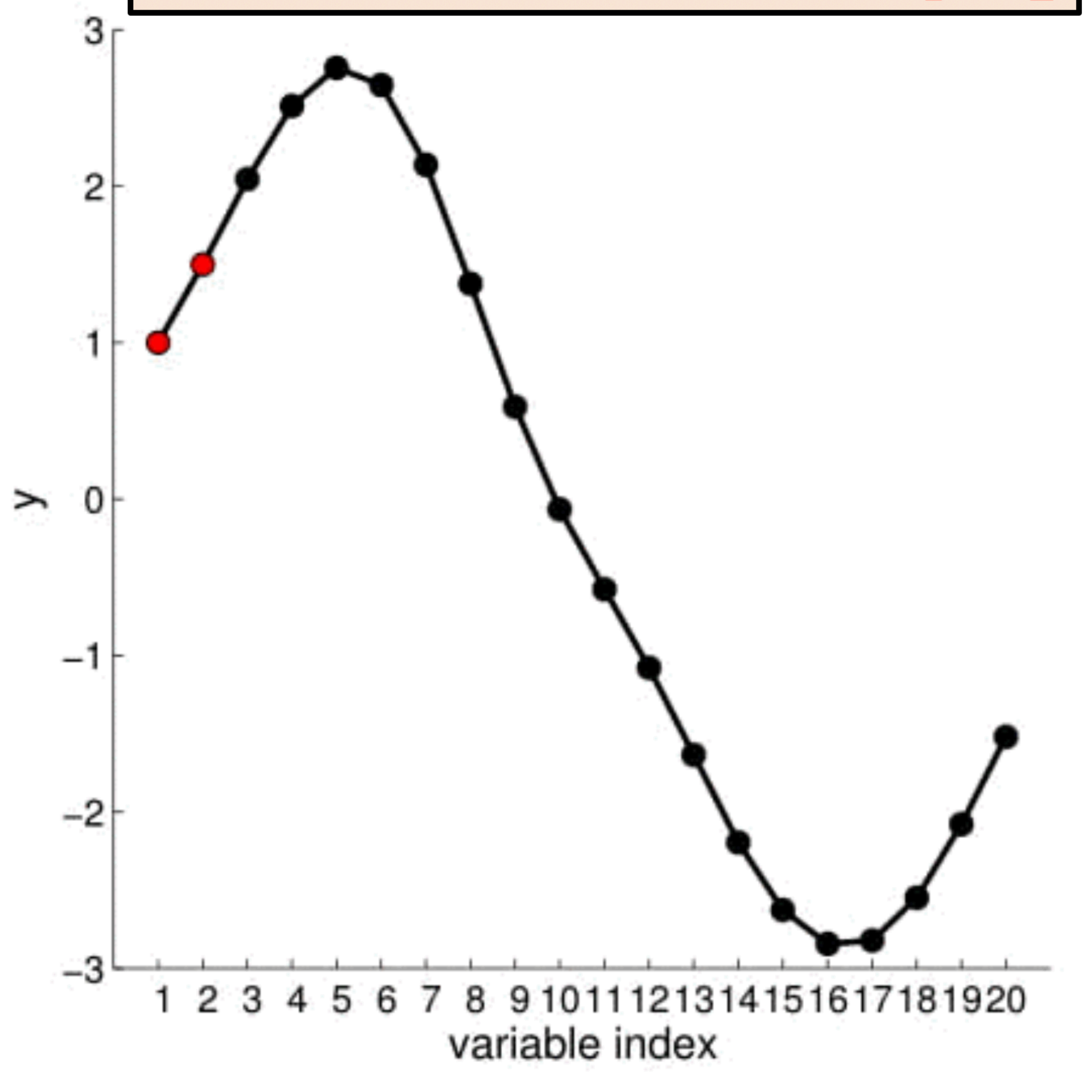

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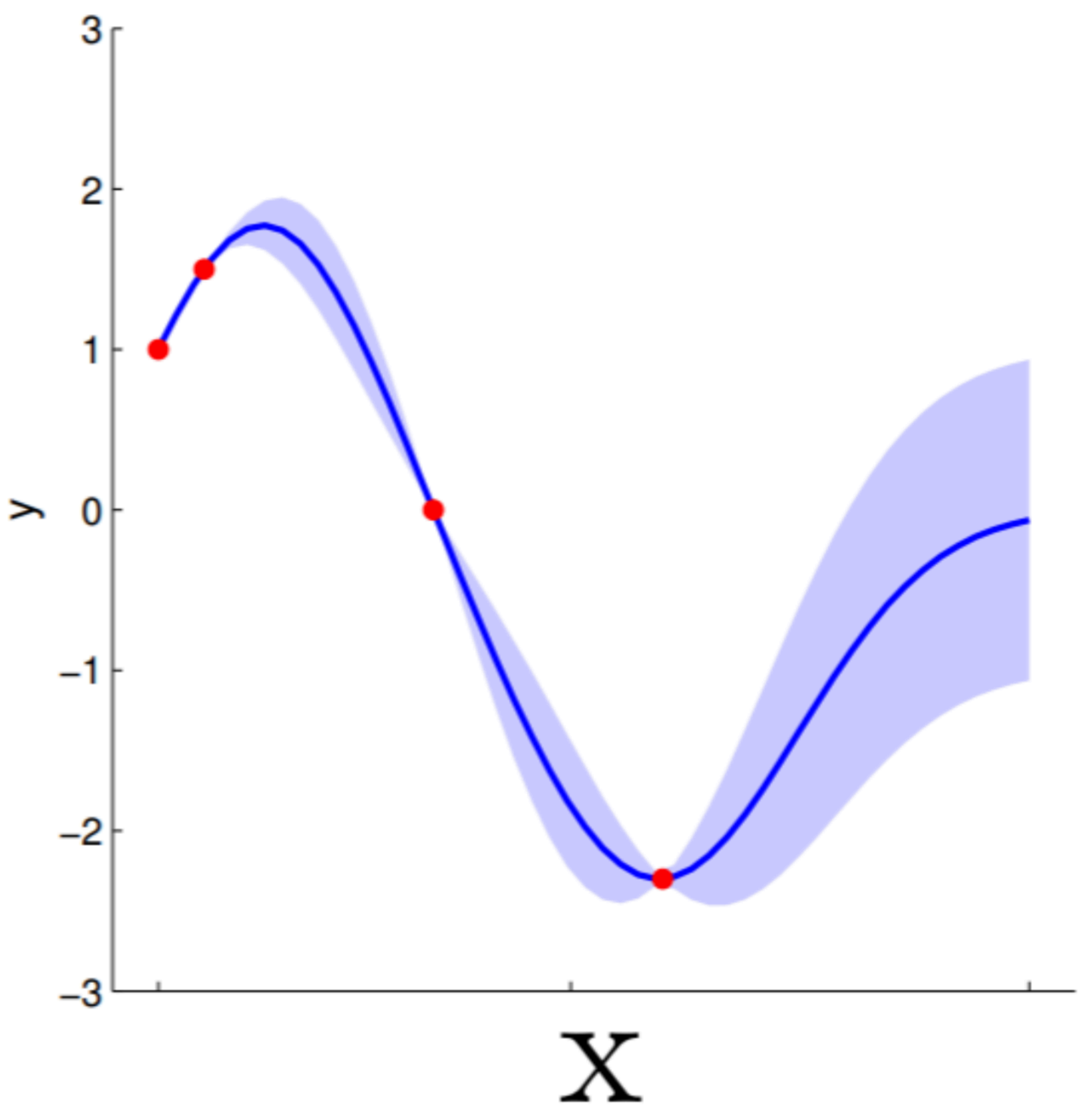

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Takeaway: A Gaussian process is a probability distribution over functions that fit a set of points. The functions have special properties determined by the covariance kernel.

Covariance kernel: central object of GP

Consider the kernel:

$$K(x_i, x_j) = \bar{c}^2 (x_i x_j)^r \exp\left(-\frac{\|x_i - x_j\|^2}{2\ell^2}\right)$$

- The kernel, for fixed values of the parameters $(\bar{c}, r, \ell)$, generates covariance matrix for the multivariate gaussian distribution.
- ℓ: correlation length between two points.
- $\bar{c}$: controls the overall scale over which the functions vary. As $\bar{c} \to 0$, GP predictions are Dirac-delta distributions.
- $r\ (\geq 0)$: generates functions whose magnitude increase with x.